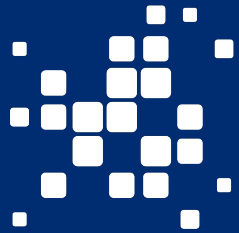
Basics of k-mers and de Bruijn graphs

Presented by Katie



wellcome sanger institute



Lex Nederbragt
@lexnederbragt

En réponse à @ctitusbrown

"Finding your way in life is like finding the genome in a De Bruijn graph: it is very easy to find *a* path, very hard to find *the* path".

Outline

- Build a k-mer counter
- **Introduction to graphs**
- Build a de Bruijn graph

Introduction to graphs

“A structure consisting of a set of objects where some pairs of the objects are in some sense ‘related’” - https://en.wikipedia.org/wiki/Graph_(discrete_mathematics)

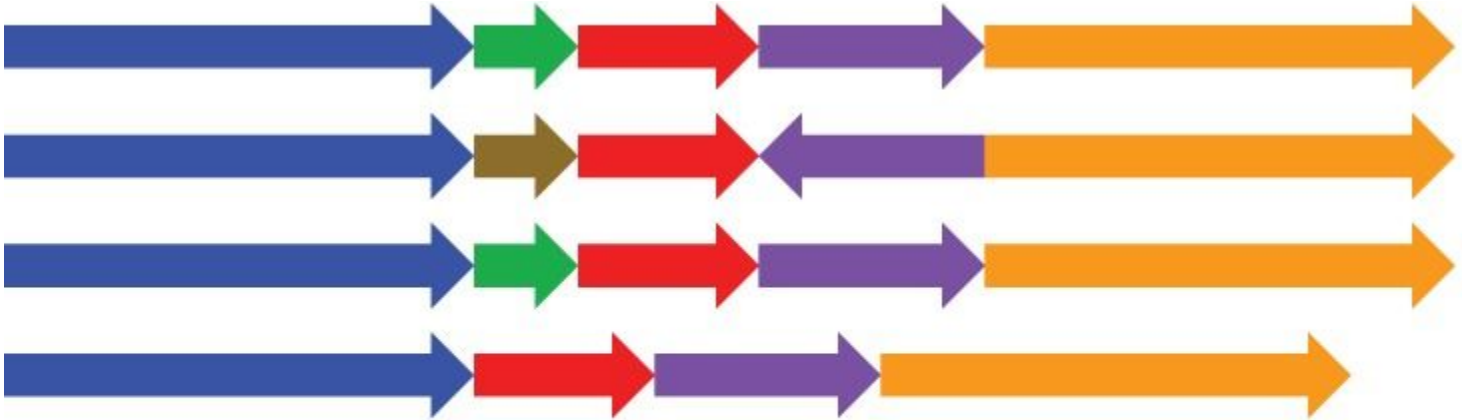
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- Genome assembly graphs
  - String graphs
  - De bruijn graphs

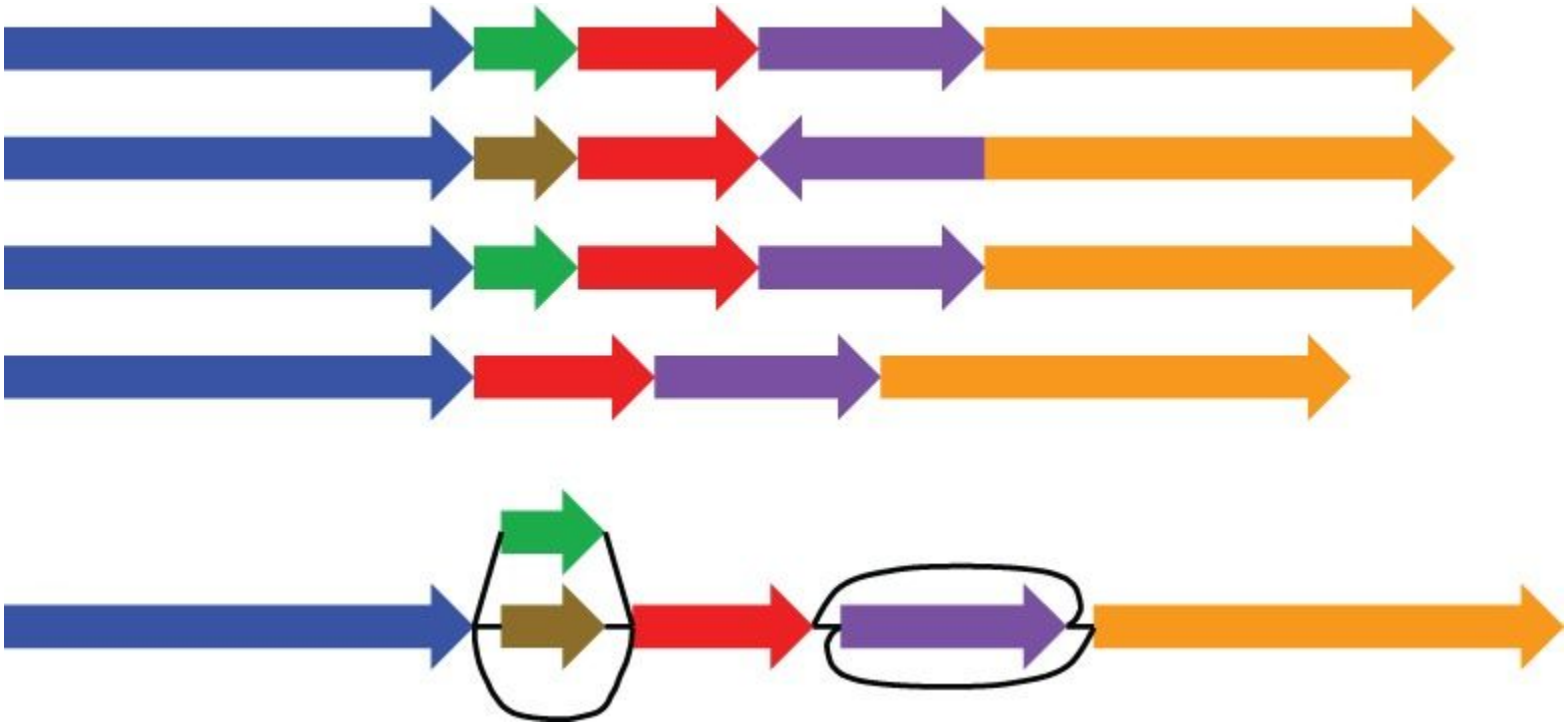
- Pangenome graphs

Introduction to graphs



https://pmc.ncbi.nlm.nih.gov/articles/PMC5411762/

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Genome assembly

- You have 1 million reads from your favorite species.

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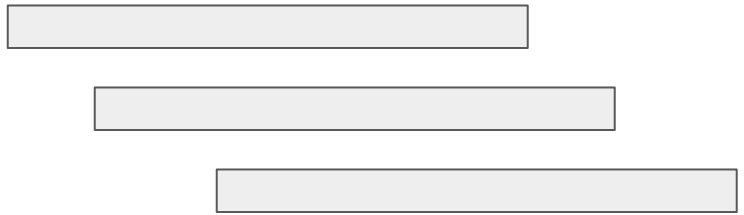
How do you reconstruct the genome from these reads?

Genome assembly

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How do you reconstruct the genome from these reads?

You *can* overlap the reads



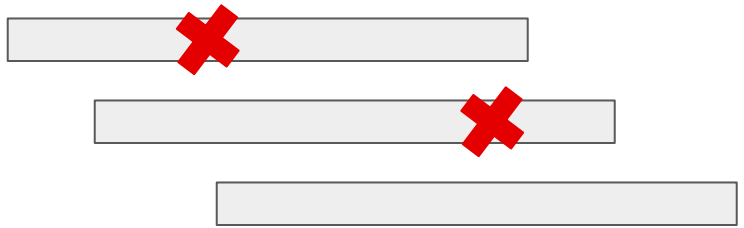
The assembly is a reconstruction of the genome that is best supported by the reads.

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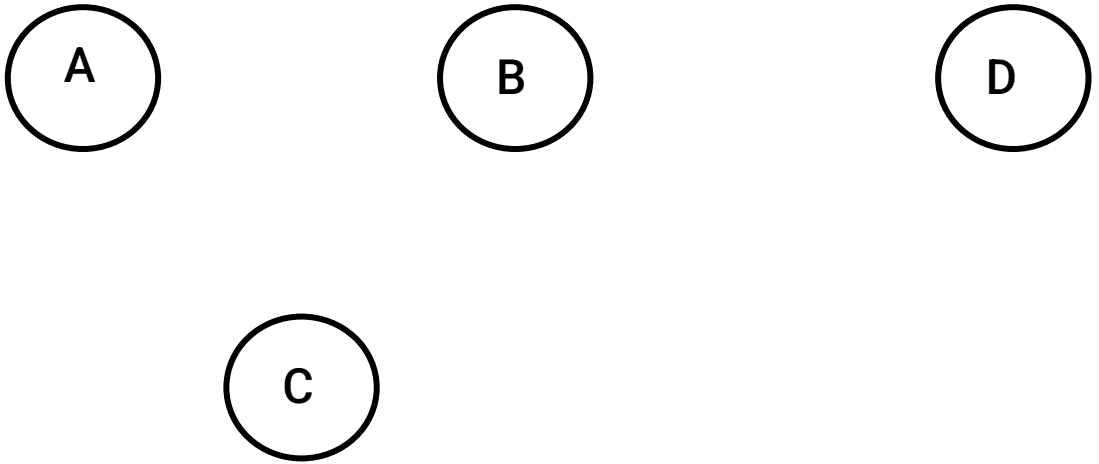
The assembly is a reconstruction of the genome that is best supported by the reads.

A graph is a way to represent relatedness between objects

“A structure consisting of a set of objects where some pairs of the objects are in some sense ‘related’” - https://en.wikipedia.org/wiki/Graph_(discrete_mathematics)

Nodes/Vertices: These are the objects

V={A,B,C,D}

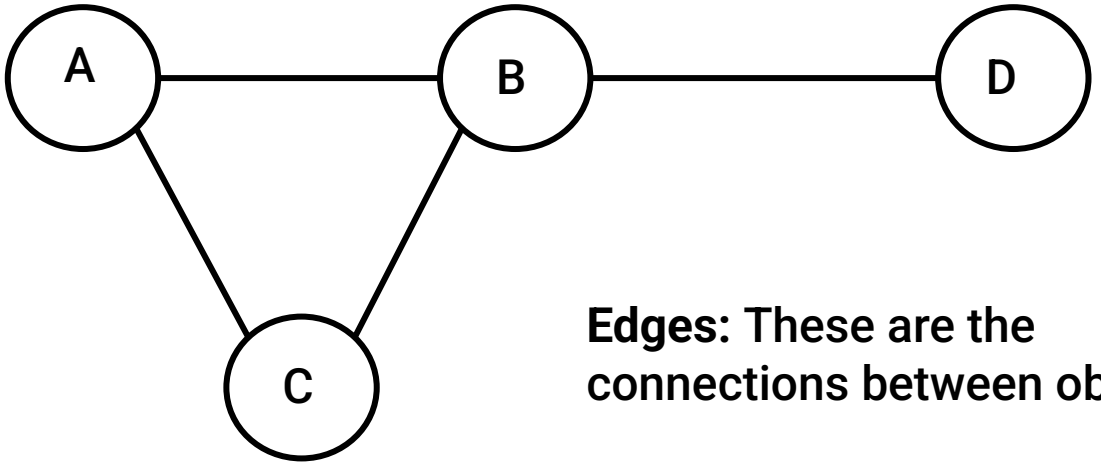


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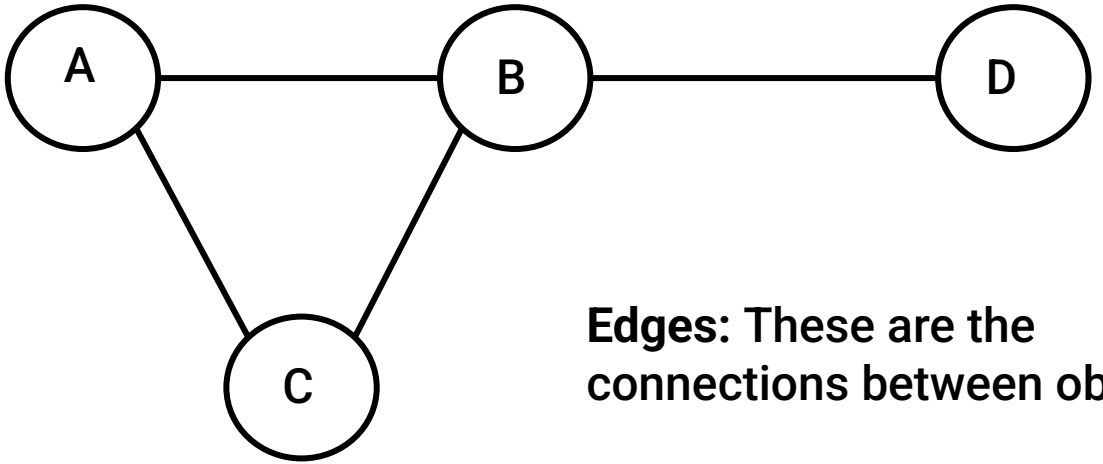
E={{A,C},{A,B},{C,B},{B,D}}

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G=(V,E)

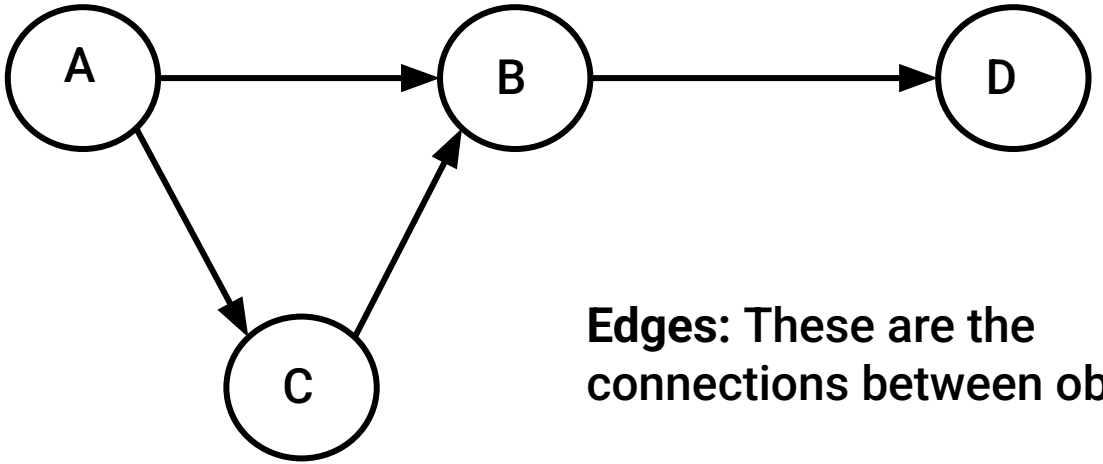
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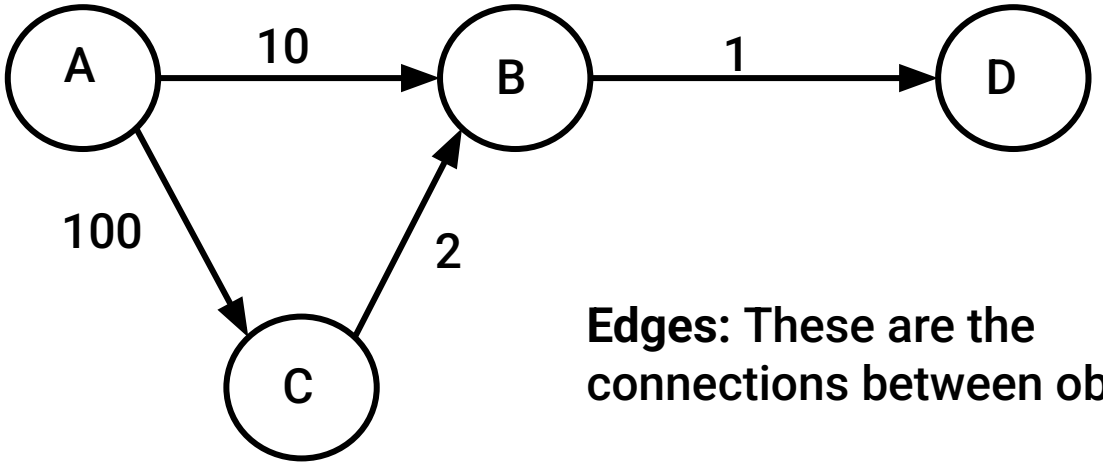
Edges: These are the connections between objects

Directed graph

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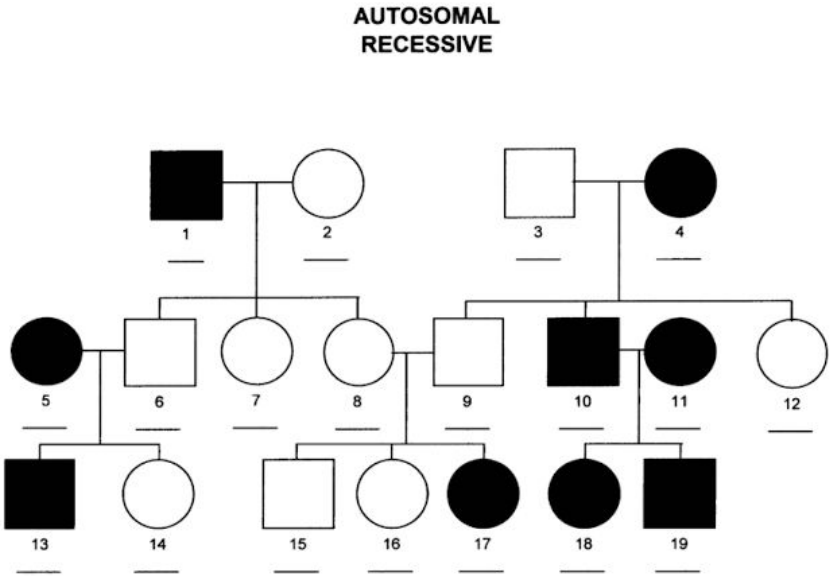


Edges: These are the connections between objects

Weighted, directed graph

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https://migrc.org/teaching-tools/genetic-inheritance-patterns/autosomal-recessive/

Introduction to graphs in genomics

“A structure consisting of a set of objects where some pairs of the objects are in some sense ‘related’” - https://en.wikipedia.org/wiki/Graph_(discrete_mathematics)

In genomics graphs are used to compactly represent many sequences

Introduction to graphs in genomics

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Two sequences, these can be represented in a graph

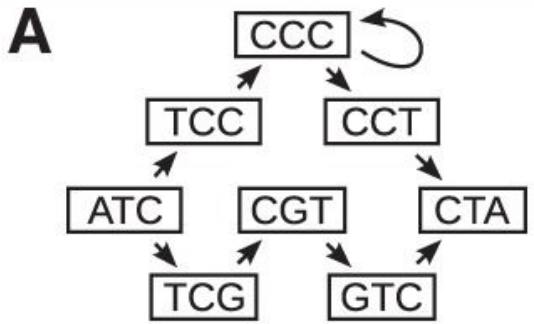
```
ATCCCCTA
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ATGTCTA
```

Introduction to graphs in genomics

ATCCCCTA ATGTCTA

De bruijn graph



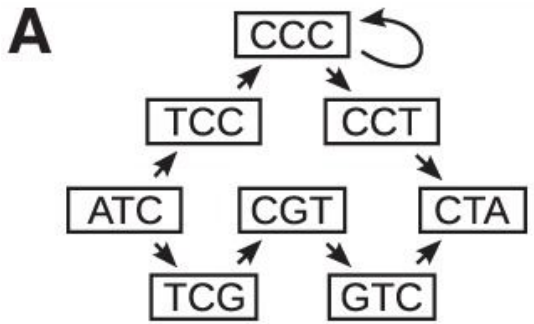
De bruijn graph: Each edge is a kmer. A walk through this graph can represent a possible genome assembly. Used in some assemblies, especially for short read data.

https://pmc.ncbi.nlm.nih.gov/articles/PMC5411762/

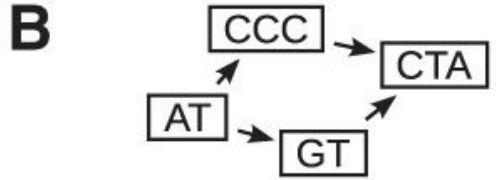
Introduction to graphs in genomics

`ATCCCCTA` `ATGTCTA`

De bruijn graph



Directed acyclic graph



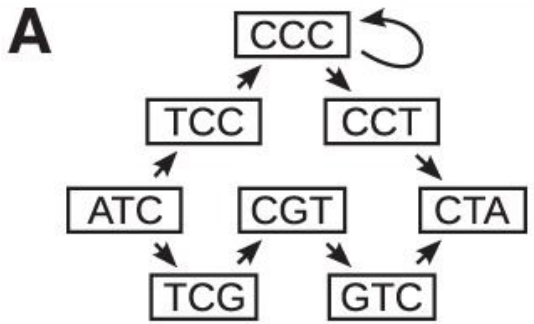
Directed acyclic graph: Made with nodes and one-way arrows that never form a closed loop. Often used for describing how genes relate to function.

https://pmc.ncbi.nlm.nih.gov/articles/PMC5411762/

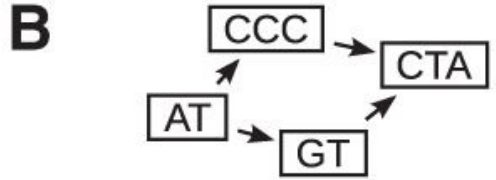
Introduction to graphs in genomics

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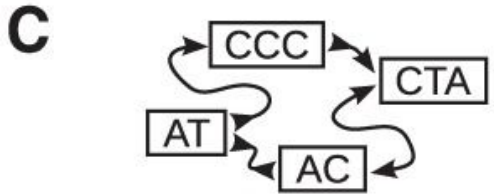
De bruijn graph



Directed acyclic graph



Sequence graph



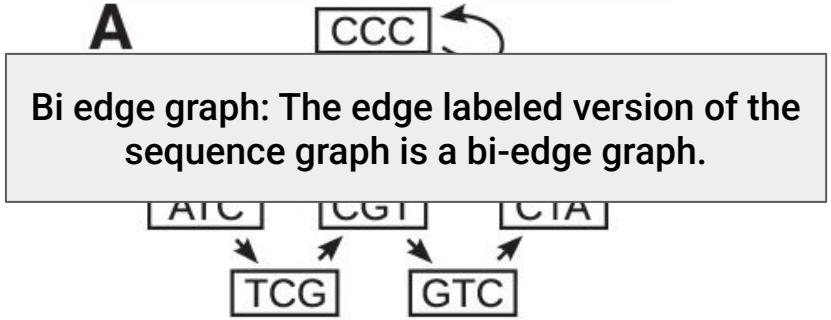
Sequence graphs: Aka, bidirected graph. Each node is labeled with a nucleotide string, and the full sequence is read out by concatenating the nodes.
This is great! But it is also hard to maintain a reference coordinate system.

https://pmc.ncbi.nlm.nih.gov/articles/PMC5411762

Introduction to graphs in genomics

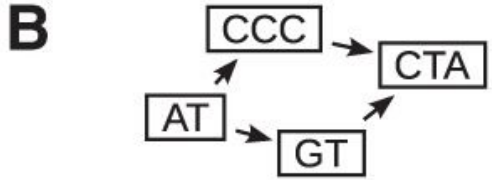
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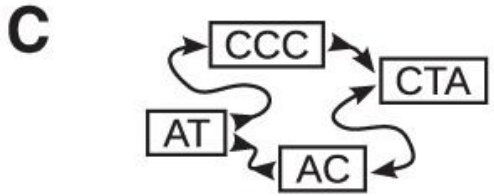


Bi edge graph: The edge labeled version of the sequence graph is a bi-edge graph.

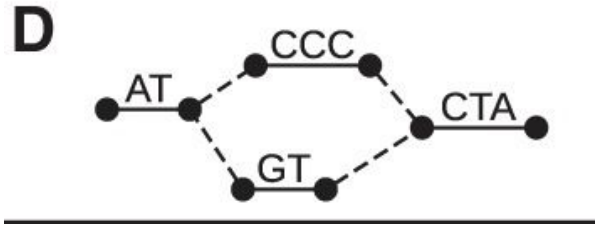
Directed acyclic graph



Sequence graph



Biedge graph

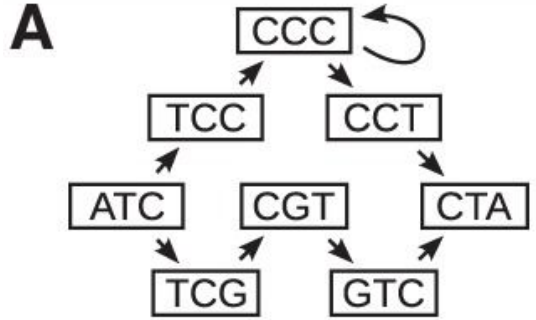


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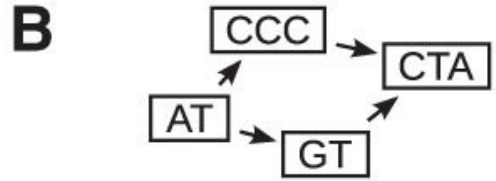
Introduction to graphs in genomics

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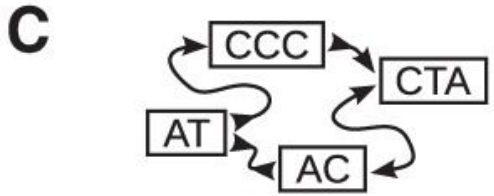
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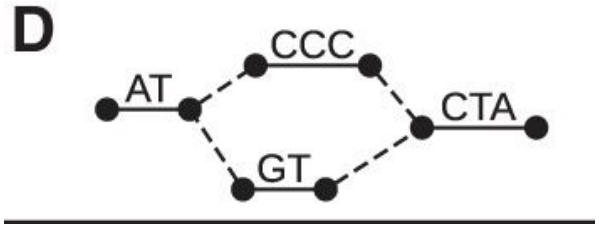
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Graphs in practice

Article | Published: 28 February 2022

Multiplex de Bruijn graphs enable genome assembly from long, high-fidelity reads

Anton Bankevich, Andrey V. Bzikadze, Mikhail Kolmogorov, Dmitry Antipov & Pavel A. Pevzner

Nature Biotechnology **40**, 1075–1081 (2022) | Cite this article

Save article

11k Accesses | **133** Citations | **127** Altmetric | Metrics

Graphs in practice

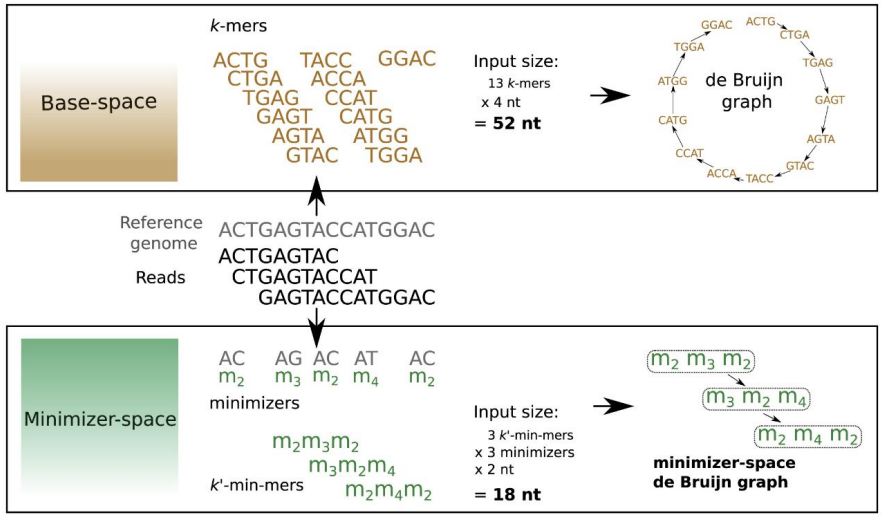
ARTICLE · Volume 12, Issue 10, P958-968.E6, October 20, 2021 · ***Open Access***

Download Full Issue

Minimizer-space de Bruijn graphs: Whole-genome assembly of long reads in minutes on a personal computer

Barış Ekim [1,2] · Bonnie Berger [1,2] · Rayan Chikhi [3,4]

A Comparison between classical and minimizer-space de Bruijn graphs (mdBG)



Graphs in practice

J Comput Biol. 2012 May;19(5):455–477. doi: 10.1089/cmb.2012.0021

SPAdes: A New Genome Assembly Algorithm and Its Applications to Single-Cell Sequencing

Anton Bankevich [1,,2], Sergey Nurk [1,,2], Dmitry Antipov [1], Alexey A Gurevich [1], Mikhail Dvorkin [1], Alexander S Kulikov [1,,3], Valery M Lesin [1], Sergey I Nikolenko [1,,3], Son Pham [4], Andrey D Prjibelski [1], Alexey V Pyshkin [1], Alexander V Sirotkin [1], Nikolay Vyahhi [1], Glenn Tesler [5], Max A Alekseyev [1,,6,✉], Pavel A Pevzner [1,,4]

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PMCID: PMC3342519 PMID: 22506599

Graphs in practice

Read 1: AATGC

Read 2: GCGGGG

Read 3: ATGCG

Write out a graph

Graphs in practice

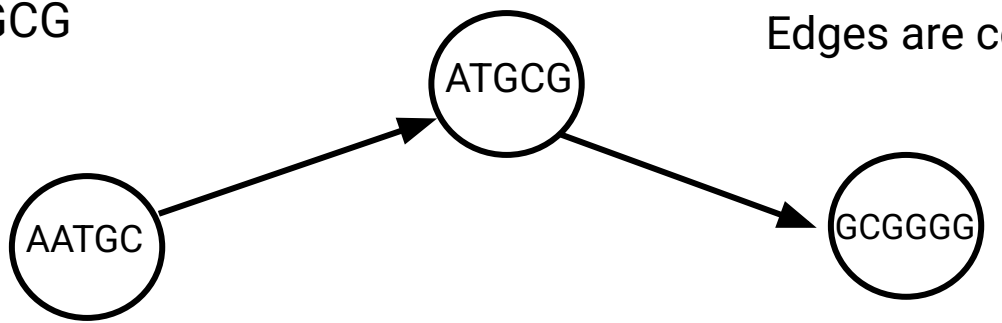
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Nodes are reads

Edges are connections



Graphs in practice

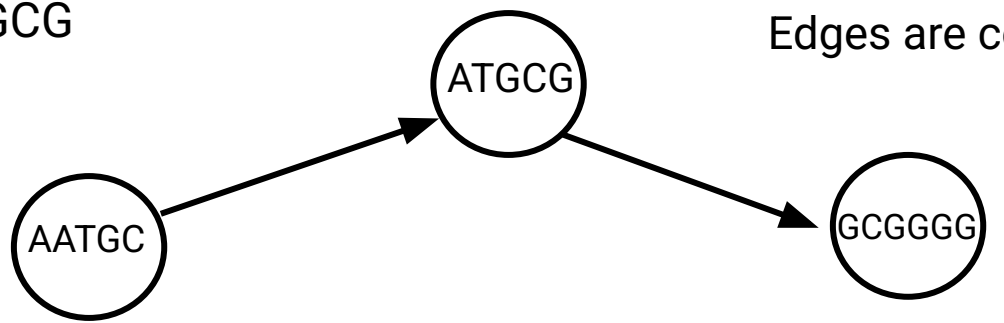
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This is fine. But there are potentially better approaches

A de Bruijn graph

- K-mers are broken up into their suffix and prefix

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- Each k-1-mer node is represented once
- The nodes are connected by k-mers that occur in the sequencing data
- A “walk” through the graph, touching every edge once, produces a possible genome sequence

Outline

- Build a k-mer counter
- Introduction to graphs
- **Build a de Bruijn graph**

Exercise:

0001110100

A de Bruijn graph uses all k-mers as edges and k-1-mers as nodes.
Draw the de Bruijn graph for the above sequence with k=3

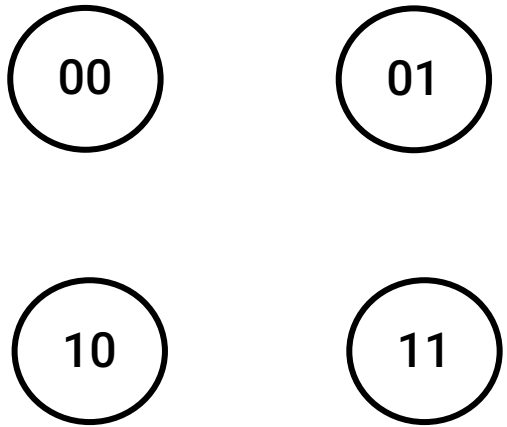
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2-mers

00
01
10
11



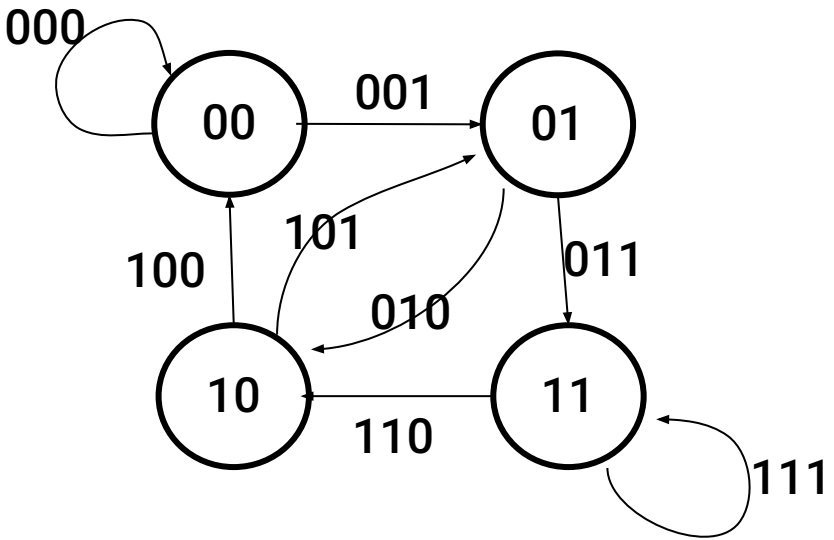
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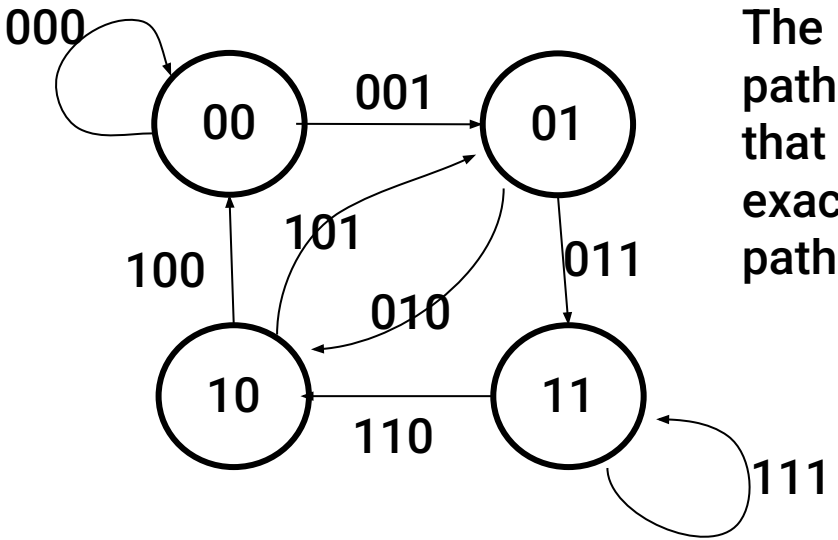
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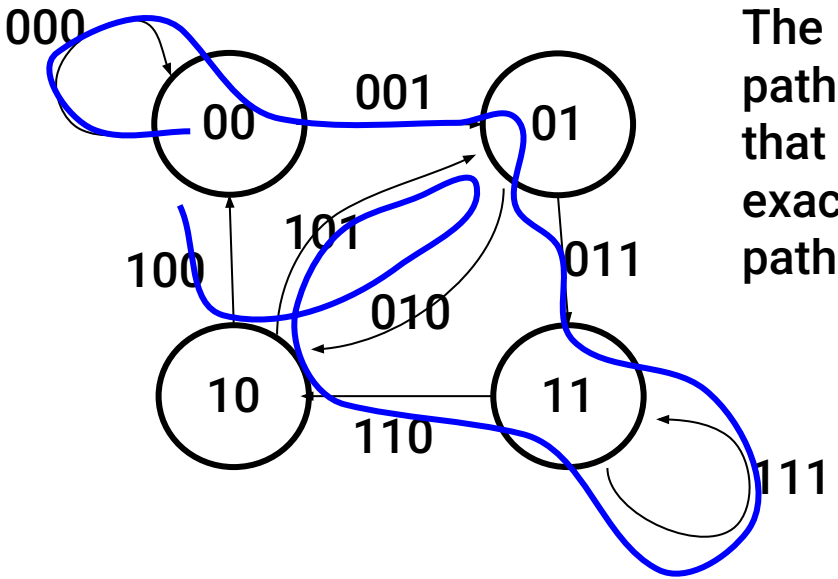
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Reads:

llcom, wellc, ellco, lcome,

K=4

De Bruijn graphs: example with larger alphabet

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llcom, wellc, ellco, lcome,

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All 4-mers:

well · ellc · lcom · come · ellc · llco · llco · lcom

De Bruijn graphs: example with larger alphabet

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llcom, wellc, ellco, lcome,

K=4

All 4-mers:

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Unique 4-mers:

well · ellc · lcom · come · llco

De Bruijn graphs: example with larger alphabet

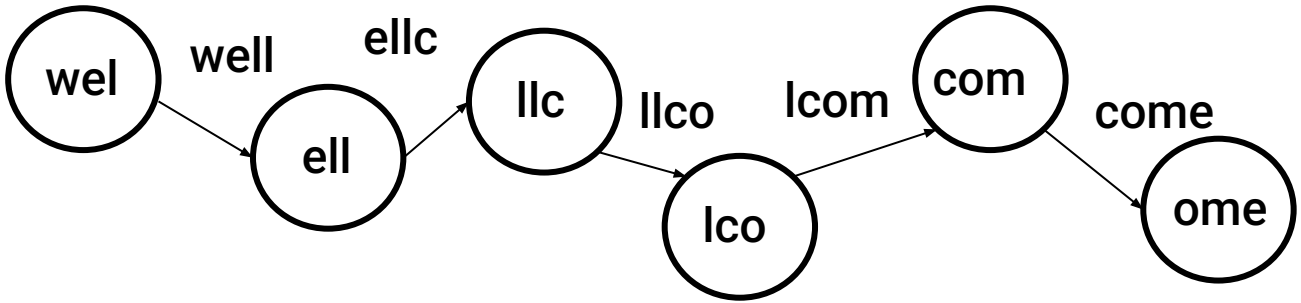
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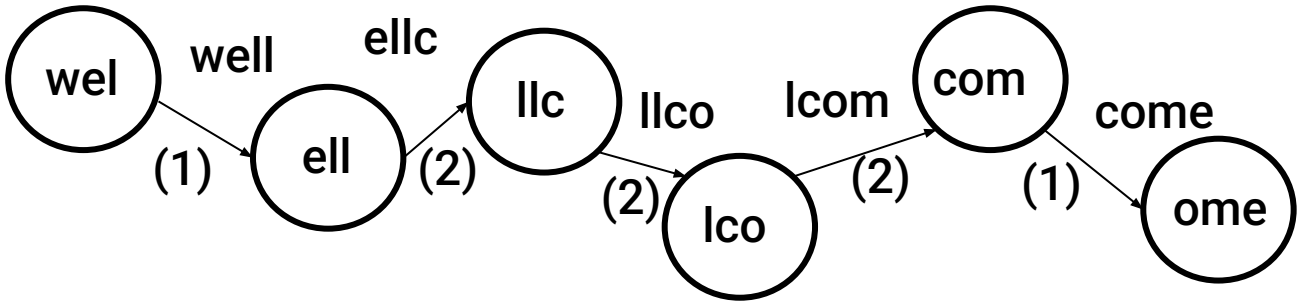
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De Bruijn graphs: example with larger alphabet

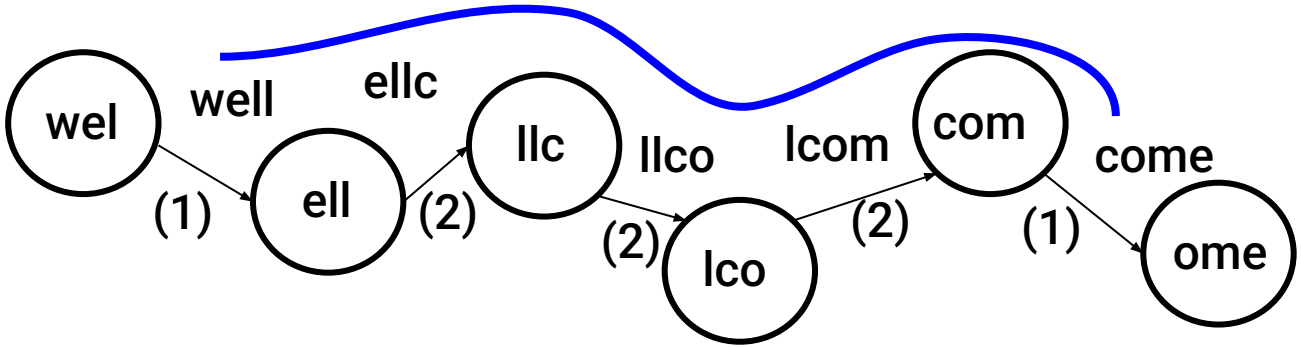
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llcom, wellc, ellco, lcome,

K=4

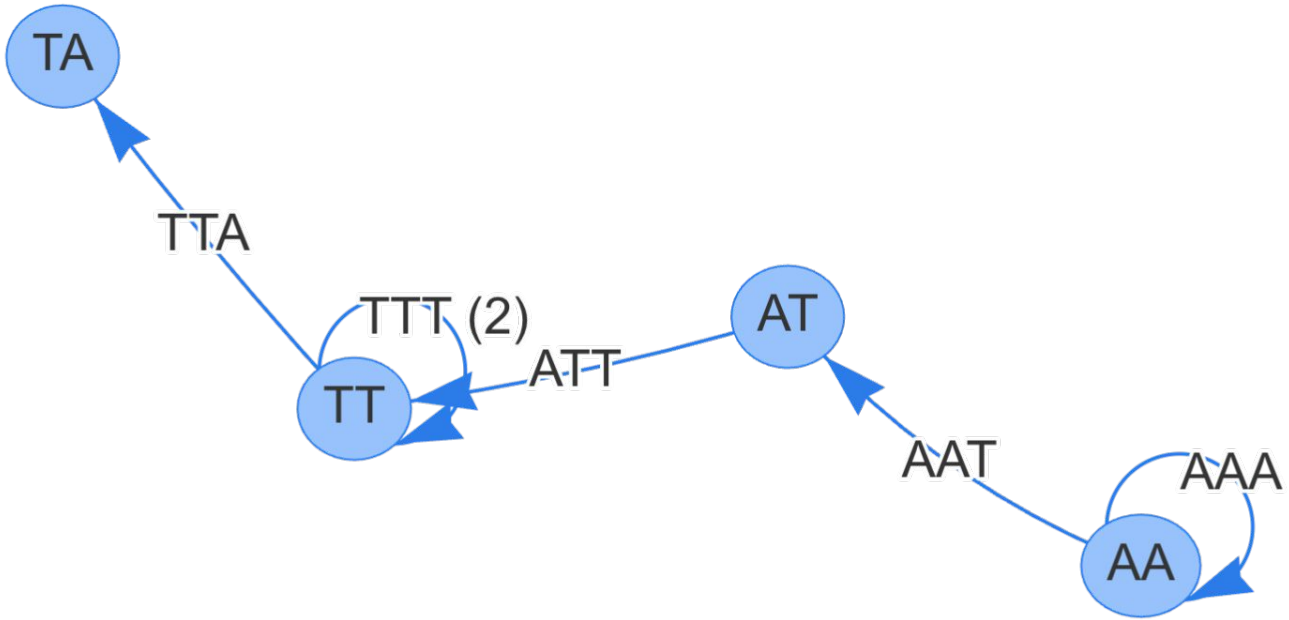
All 4-mers:

well · ellc · lcom · come · ellc · llco · llco · lcom



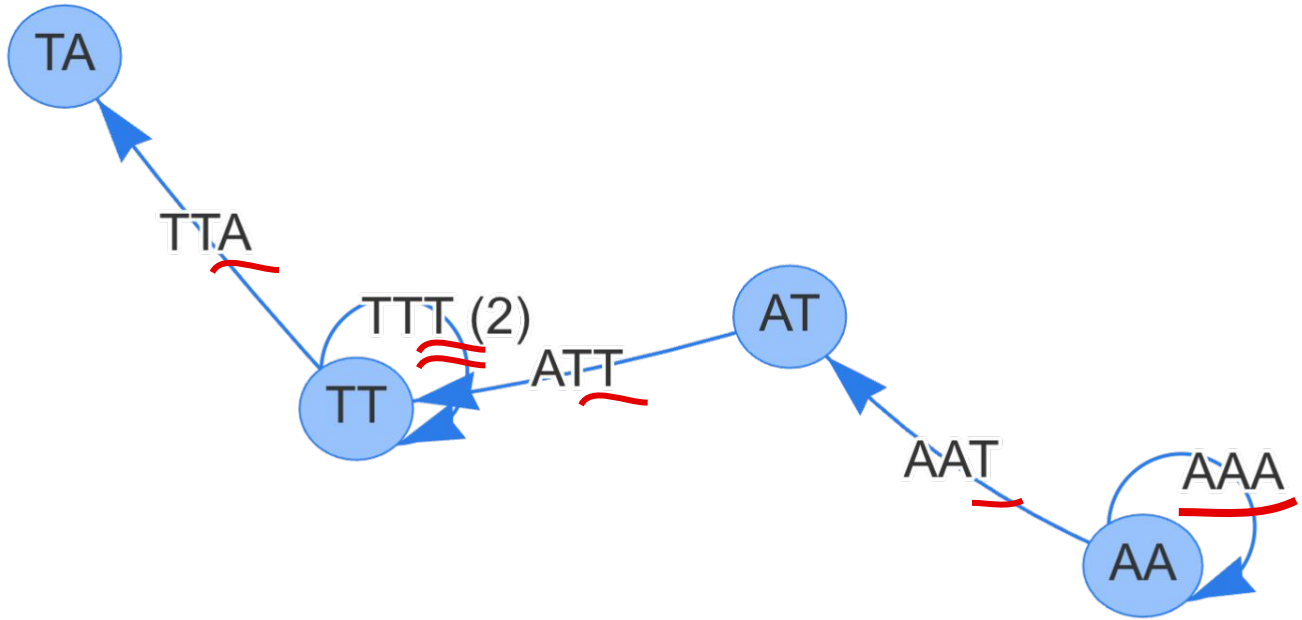
De Bruijn graphs: example

From the graph, what is the genome?



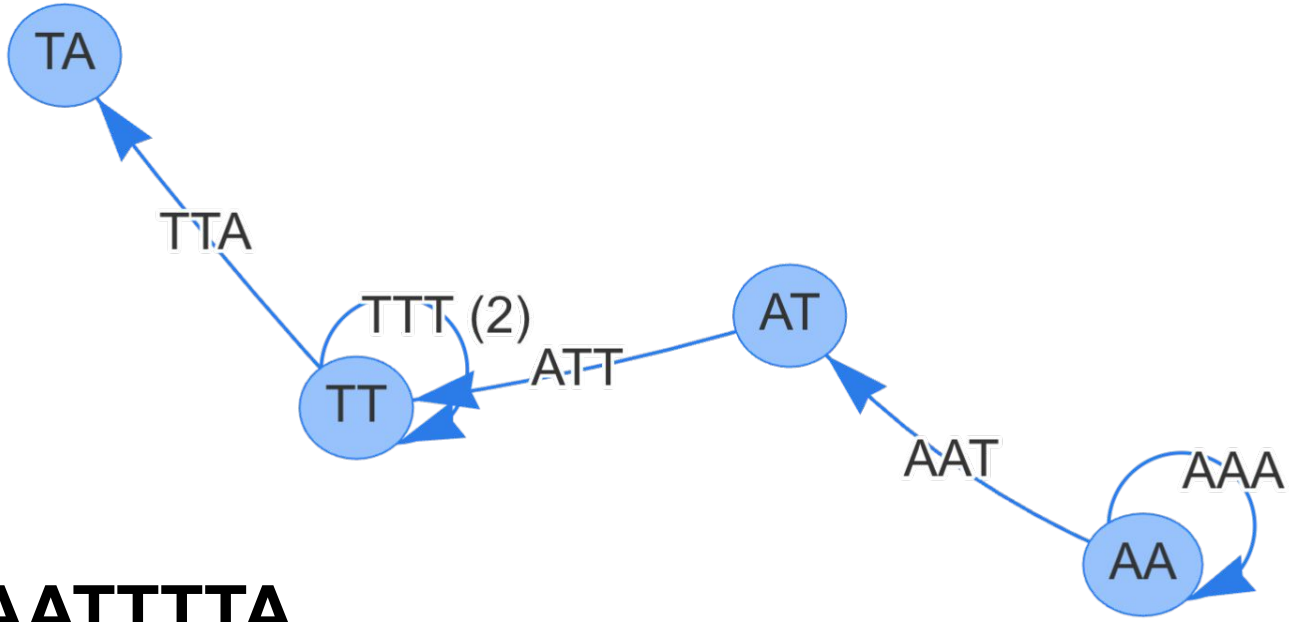
De Bruijn graphs: example

From the graph, what is the genome?



De Bruijn graphs: example

From the graph, what is the genome?



AAATTTTA

De Bruijn graphs: example with repeats

Reads:

s_and, dogs_, ogs_a, d_cat, cats_, ats_a, gs_an, s_and, and_c, d_cat, nd_do, _dogs, d_dog, _and_

K=3

Construct the graph

De Bruijn graphs: example with repeats

Reads:

s_and, dogs_, ogs_a, d_cat, cats_, ats_a, gs_an, s_and, and_c, d_cat, nd_do, _dogs, d_dog, _and_

K=3

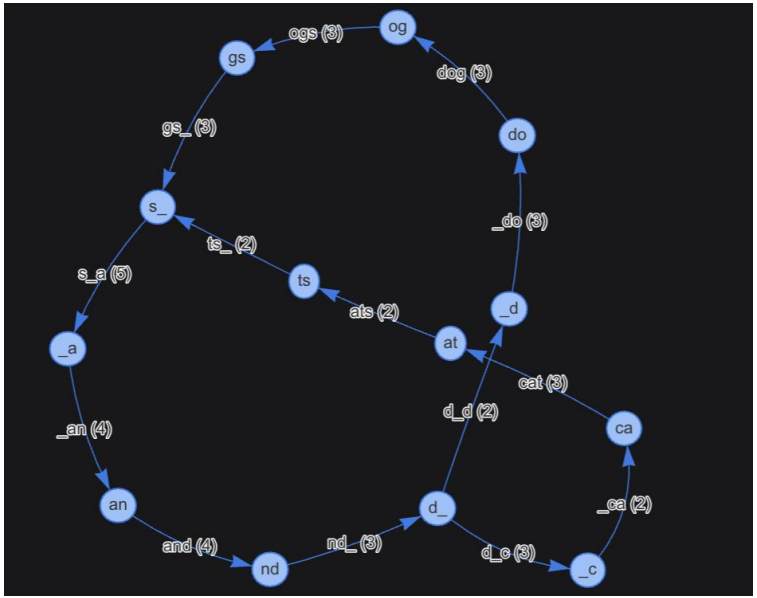
dog · ogs · gs_ · gs_ · s_a · _an · s_a · _an · and · ogs · gs_ · s_a · d_c · _ca · cat · cat · ats · ts_ · ats · ts_ · s_a · and · nd_ · d_c · d_c · _ca · cat · s_a · _an · and · nd_ · d_d · _do · _do · dog · ogs · d_d · _do · dog · _an · and · nd_

De Bruijn graphs: example with repeats

Reads:

s_and, dogs_, ogs_a, d_cat, cats_, ats_a, gs_an, s_and, and_c, d_cat, nd_do, _dogs, d_dog, _and_

K=3

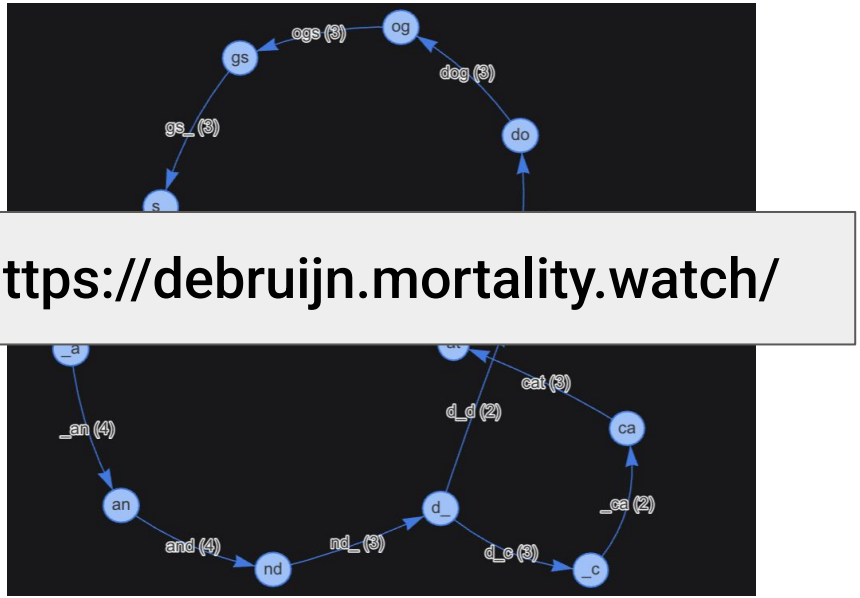


De Bruijn graphs: example with repeats

Reads:

s_and, dogs_, ogs_a, d_cat, cats_, ats_a, gs_an, s_and, and_c, d_cat, nd_do, _dogs, d_dog, _and_

K=3



De Bruijn graphs: example

1. Download all of the reads from the de_bruij_graph_reads folder in the course_data
2. Go to this URL: https://debruijn.mortality.watch/

   This is also in the course github

https://debruijn.mortality.watch/

De Bruijn graphs: example

Use the test reads given (read length 4-10) in the provided website.

Adjust k for each of the read lengths.

What is the impact of read length on de bruijn graph construction?

What is the impact of k on the graph?

What do you think the original sequence was?

https://debruijn.mortality.watch/

De Bruijn graphs: example

```
python simulate_reads.py \
    --length 100 \
    --coverage 2 \
    --read-length 5 \
    --error-rate 0.01 \
    --heterozygosity 0.001 \
    --seed 42
```

1. Adjust the coverage - what is the impact of coverage on interpretability?

https://debruijn.mortality.watch/

De Bruijn graphs: example

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    --length 100 \
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1. Adjust the coverage - what is the impact of coverage on interpretability?
2. Add and remove errors. How do errors impact graph topology

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De Bruijn graphs: example

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2. Add and remove errors. How do errors impact graph topology
3. Adjust the read length

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    --seed 42
```

1. Adjust the coverage - what is the impact of coverage on interpretability?
2. Add and remove errors. How do errors impact graph topology
3. Adjust the read length
4. Adjust sequence length
5. Adjust K

- How do each of these parameters impact the graph?
- What different graph topologies did you see?
- Did anyone change multiple parameters at a time?
- How might all of this impact genome assembly?

Impact of k on de bruijn graphs

Shorter k

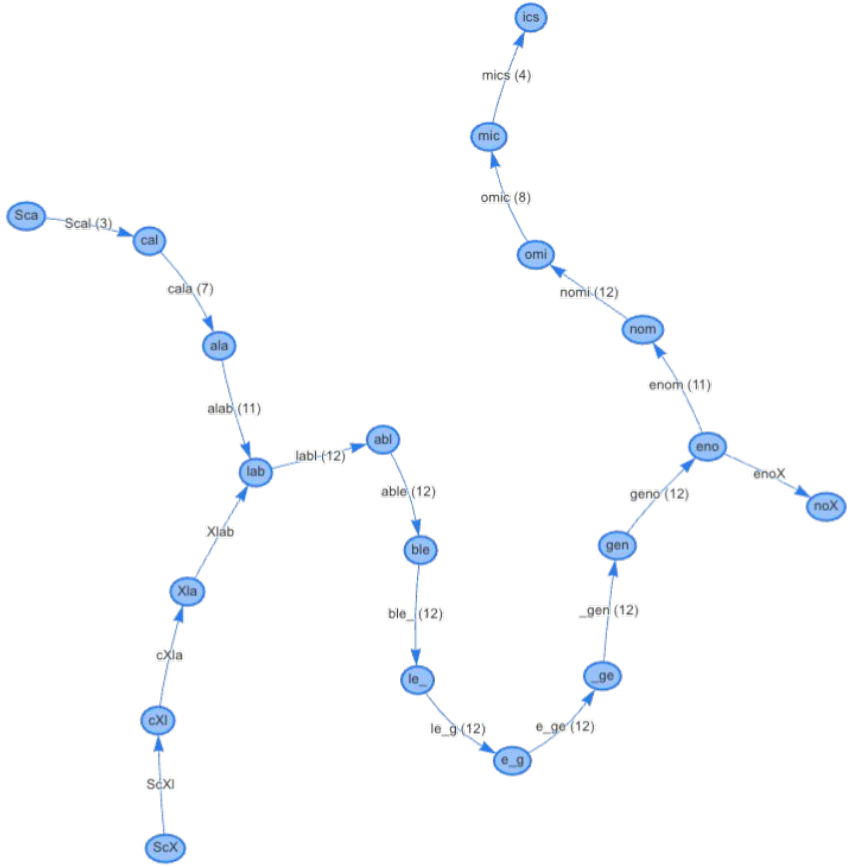
- Higher k coverage
- Repeats are less resolvable
- More robust to errors

Longer k

- Very helpful with repeats
- Sensitive to errors
- Lower k coverage

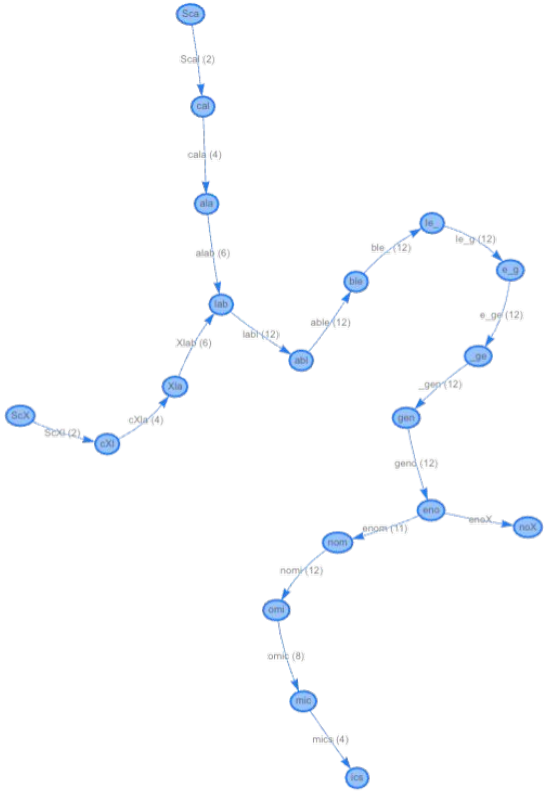
De Bruijn graphs: examples

- What do these branches indicate? And what evidence supports your theory?



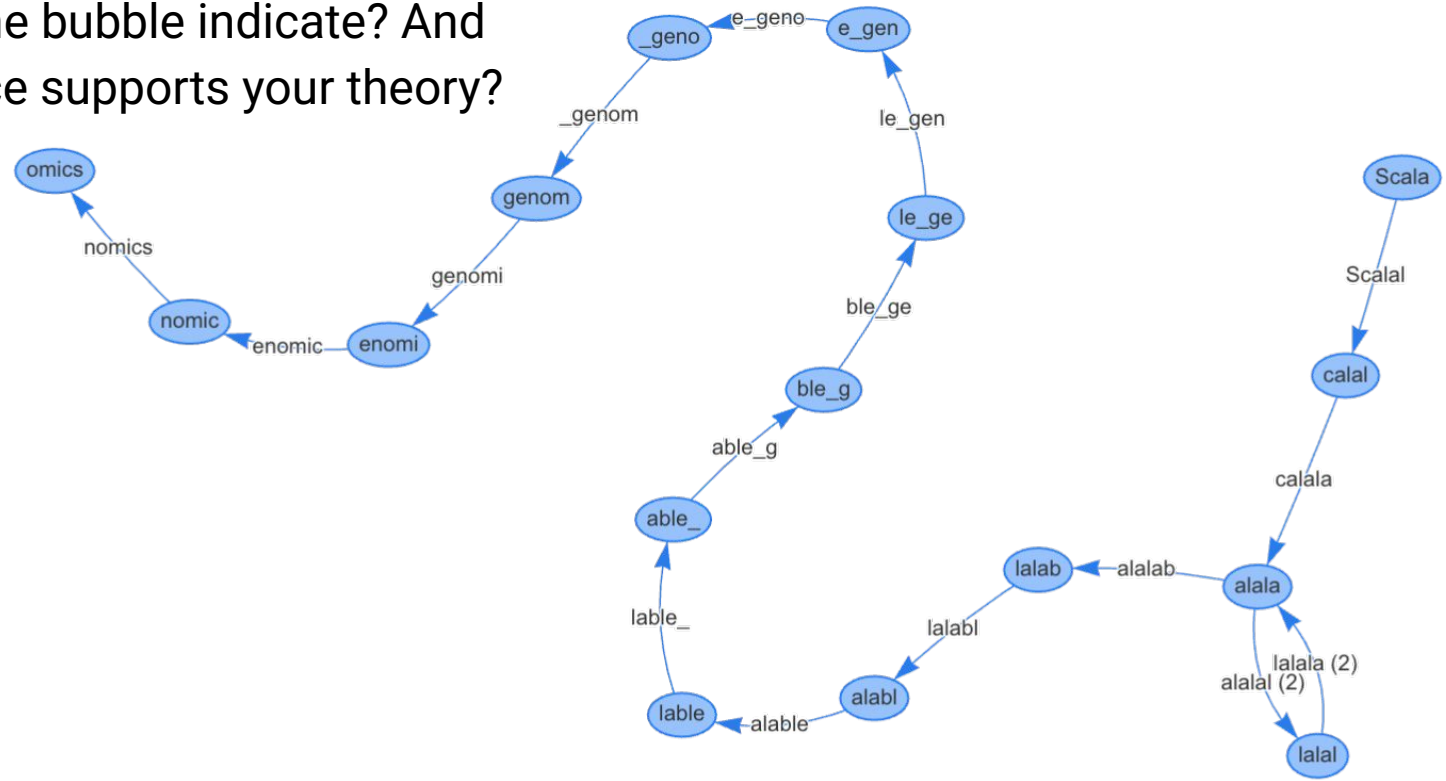
De Bruijn graphs: examples

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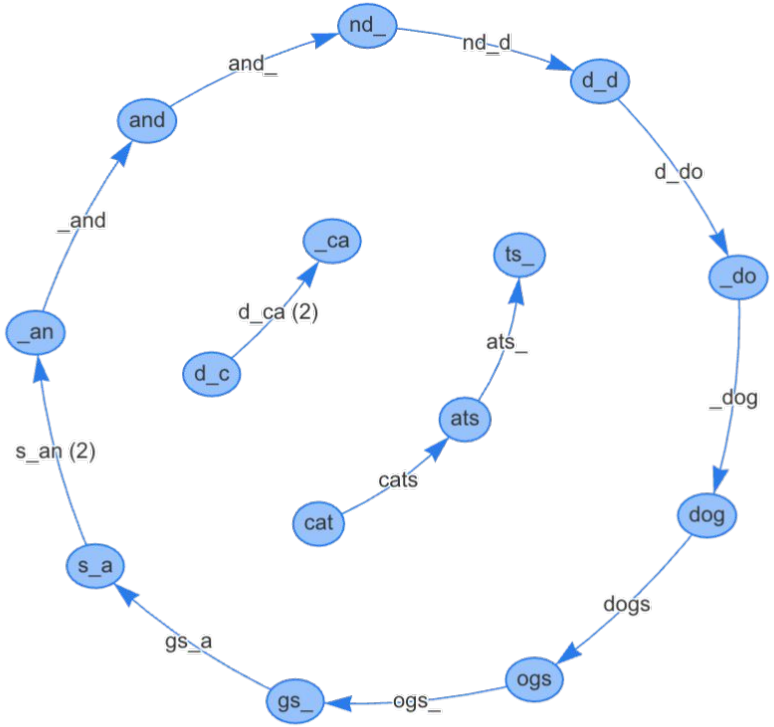
De Bruijn graphs: examples

- What does the bubble indicate? And what evidence supports your theory?



De Bruijn graphs: examples

- Why are there unconnected components?

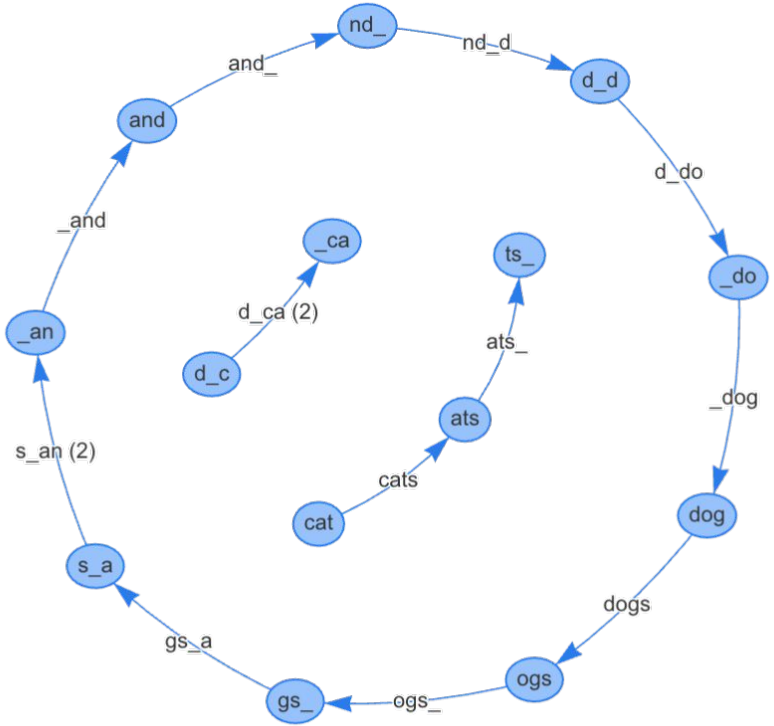


De Bruijn graphs: examples

- Why are there unconnected components?

Kmers: dogs
gs_a
s_an
ogs_
d_ca
cats
ats_
and_
d_ca
s_an
nd_d
_dog
d_do
_and

dogs_and_cats_and_cats_and_dogs
dogs d_ca and_
gs_a d_ca _and cats
s_an s_an ats_ nd_d
ogs_ d_do
_dog



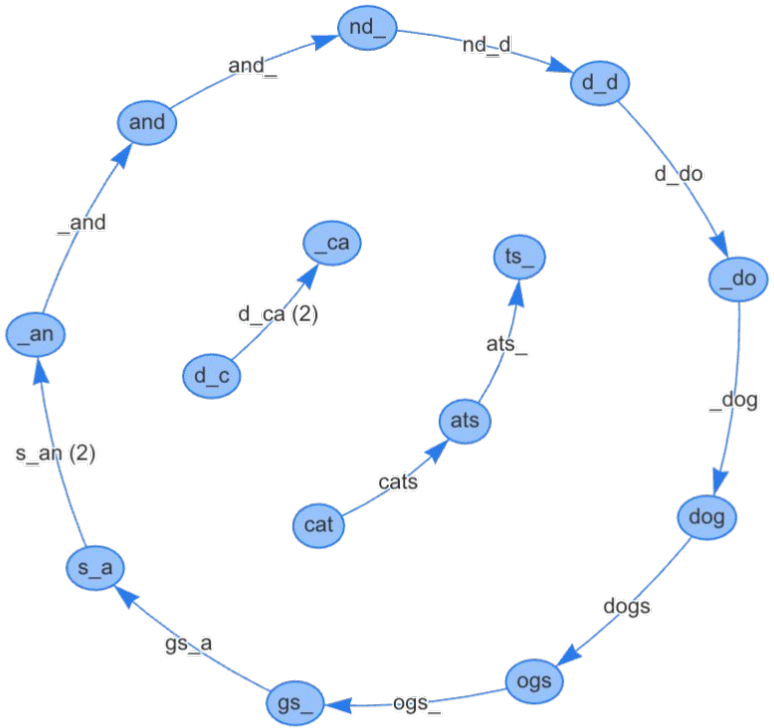
De Bruijn graphs: examples

- Why are there unconnected components?

Kmers:

dogs
gs_a
s_an
ogs_
d_ca
cats
ats_
and_
d_ca
s_an
nd_d
_dog
d_do
_and

dogs_and_cats_and_cats_and_dogs
dogs d_ca and_
gs_a d_ca _and cats
s_an s_an ats_ nd_d
ogs_ d_do
_dog



De Bruijn graphs: examples

- Why are there unconnected components?

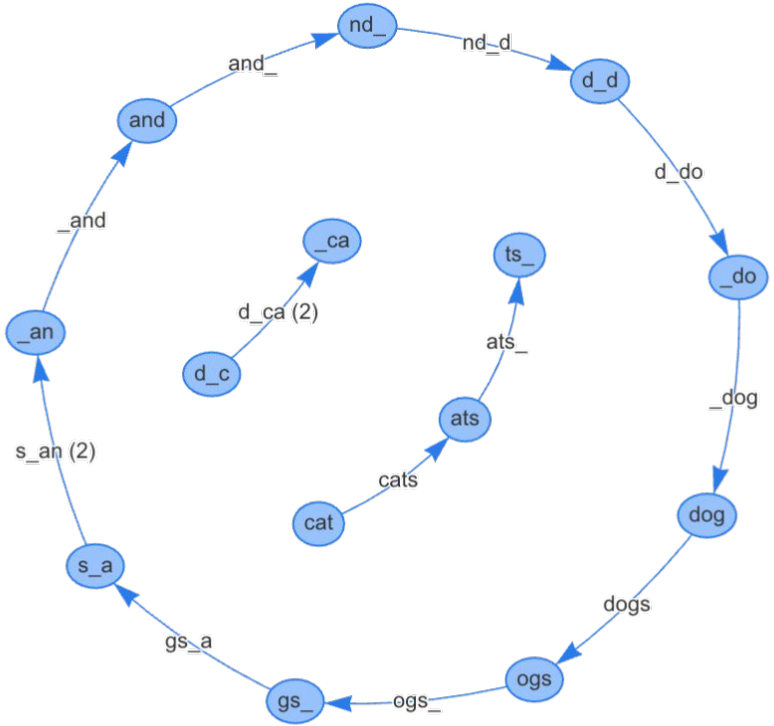
Kmers:

dogs
gs_a
s_an
ogs_
d_ca
cats
ats_
and_
d_ca
s_an
nd_d
_dog
d_do
_and

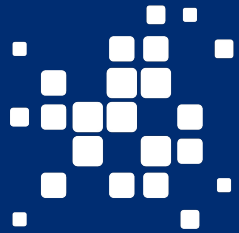
dogs_and_cats_and_cats_and_dogs

dogs d_ca and_
gs_a d_ca _and cats nd_d
s_an s_an ats_ d_do
ogs_ _dog

What might fix this?

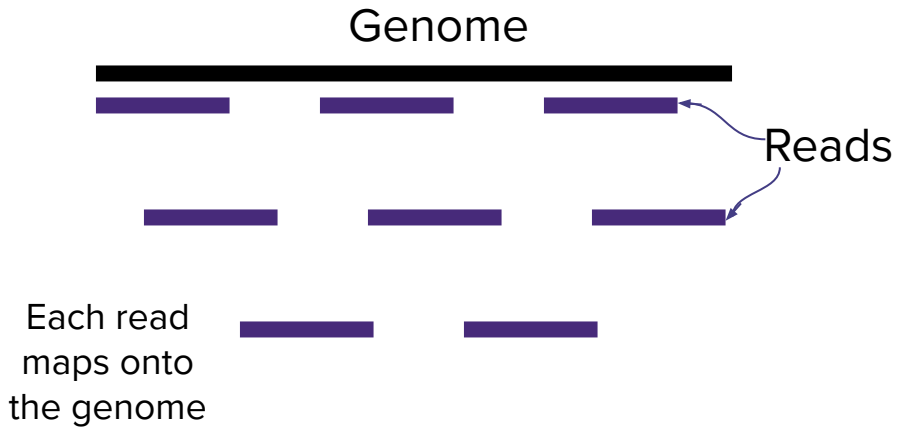


Questions?

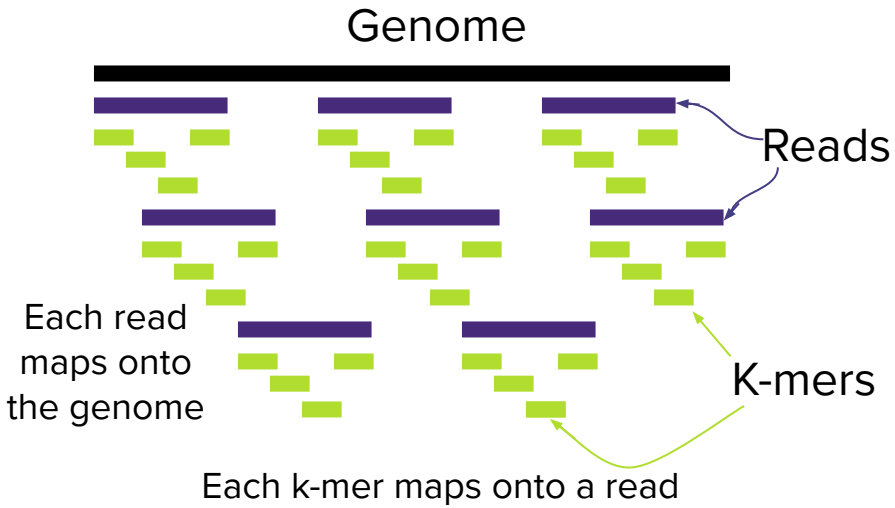


wellcome sanger institute

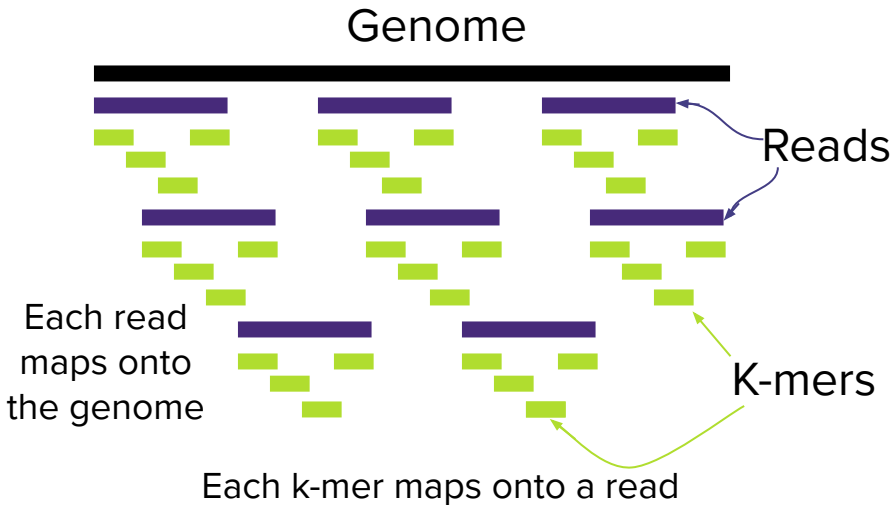
Genome k-mers vs read k-mers



Genome k-mers vs read k-mers



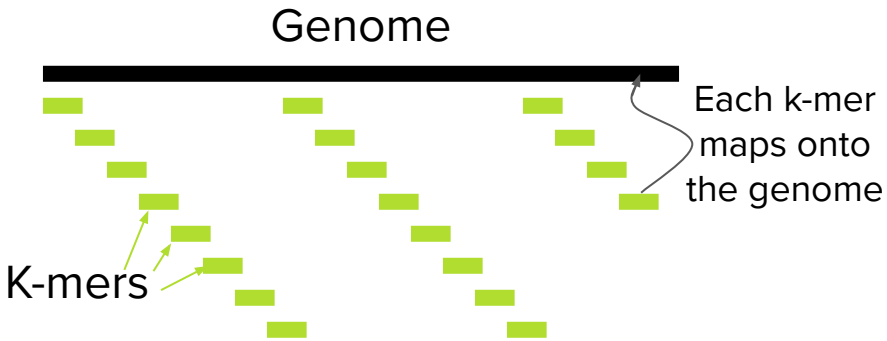
Genome k-mers vs read k-mers

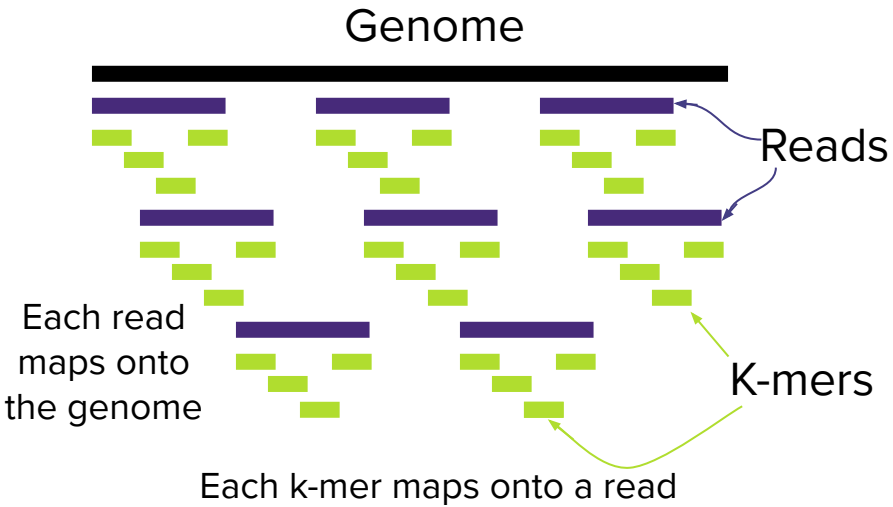


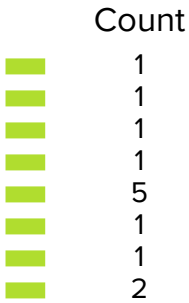
Count
3
3
3
3
15
3
3
10

Higher counts reflect **higher coverage** of that region and/or more **repetitive sequence**

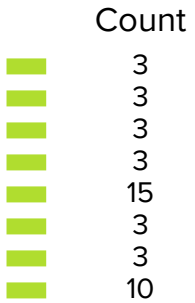
Genome k-mers vs read k-mers







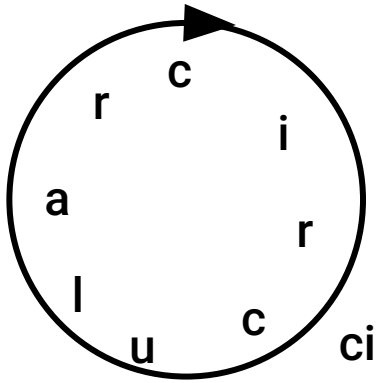
Higher counts reflect **repetitive sequence** (occurs multiple times throughout the genome)



Higher counts reflect **higher coverage** of that region and/or more **repetitive sequence**

Exercise:

Find the shortest circular superstring that contains all possible substrings of length k in a given alphabet.



circular

Alphabet = {0,1}
Superstring longer string.
Substring is a string in the superstring

Exercise:

All possible 3-mers of alphabet {0,1}
2^3 = 8

000
001
010
100
110
101
011
111

Exercise:

Find the shortest circular superstring that contains all possible substrings of length k in a given alphabet.

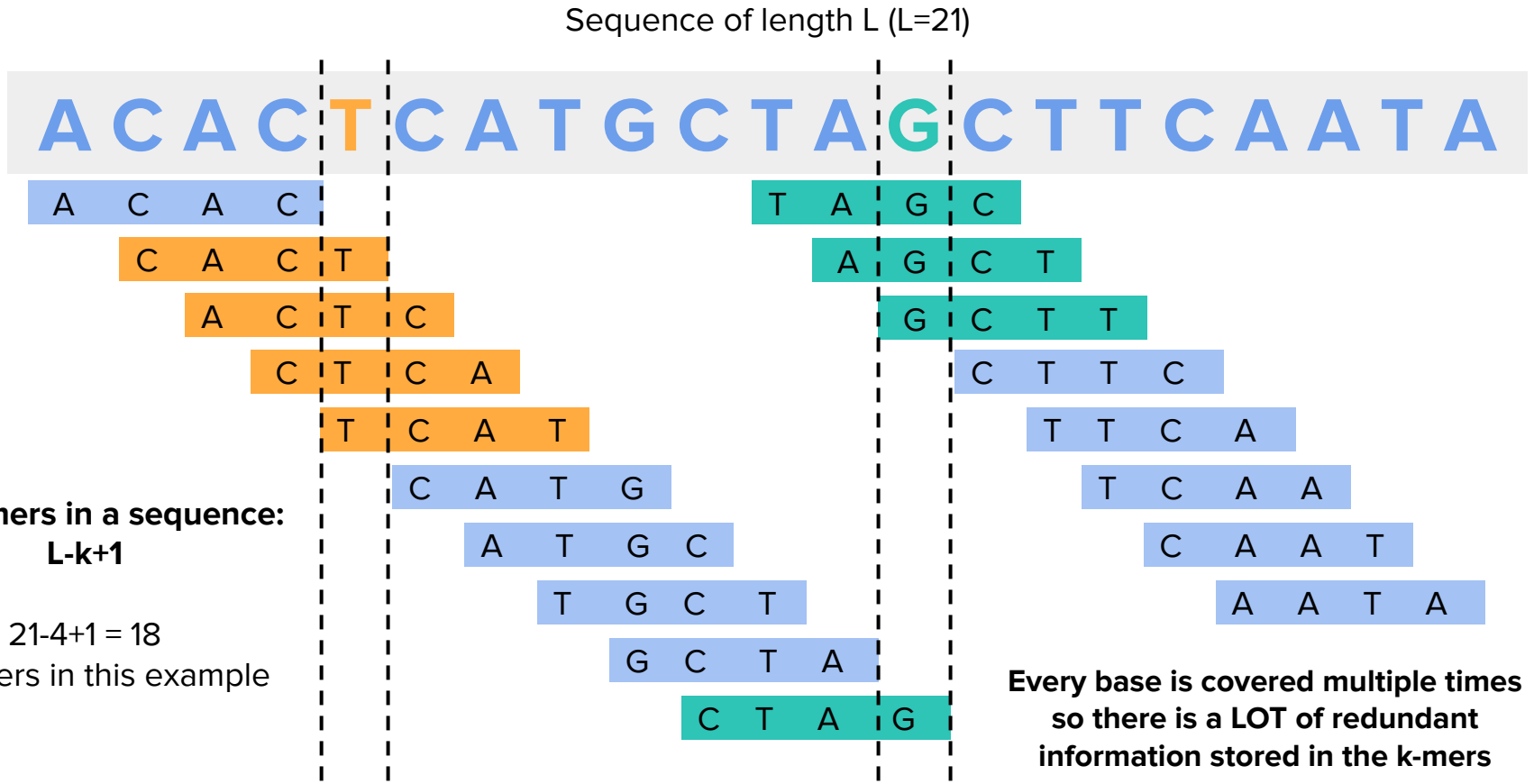
0001110100

All 3-mers are contained exactly once. So it is the shortest possible superstring that contains all substrings of length k

Outline

- Build a k-mer counter
- Introduction to graphs
- Build a de Bruijn graph

A k-mer is a substring of length k



of k-mers in a sequence: L-k+1

21-4+1 = 18
18 4-mers in this example

Every base is covered multiple times so there is a LOT of redundant information stored in the k-mers

Factors that influence read k-mer counts

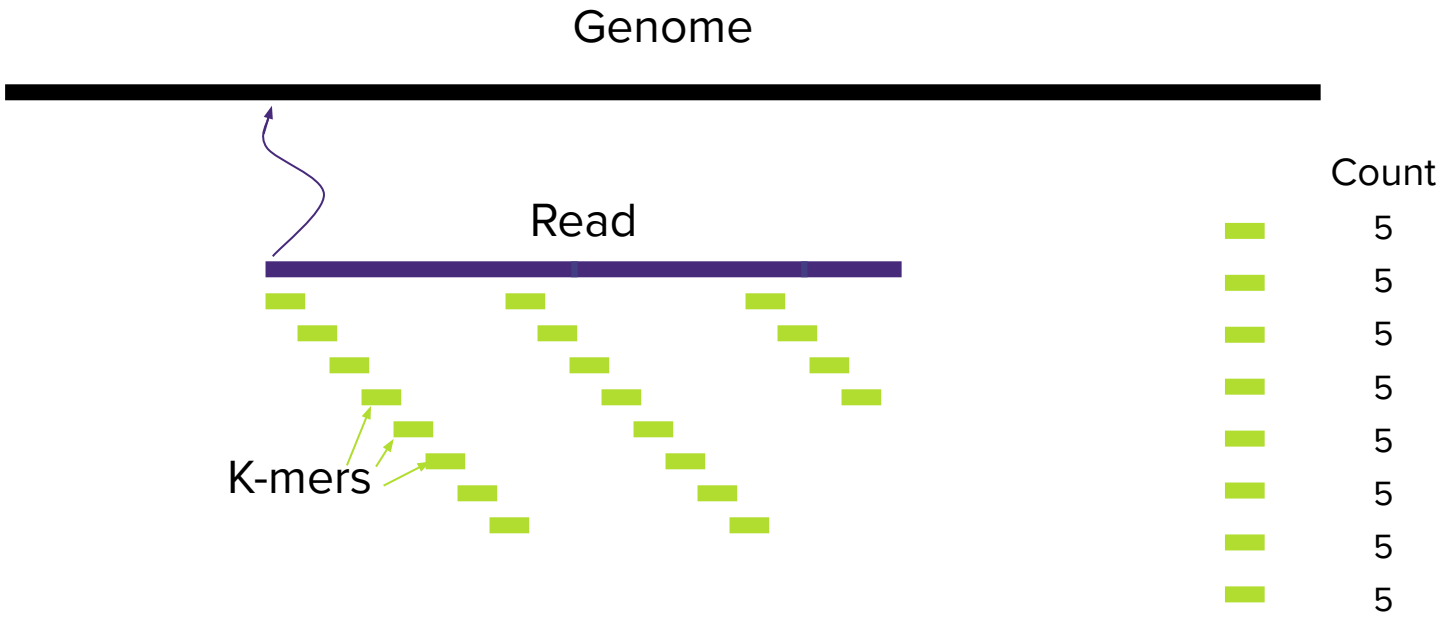
- Read **error** level
- Genome **coverage**
- **Read length**
- **Heterozygosity** of the genome
- Genome **ploidy**

Factors that influence read k-mer counts

- Read **error** level
- Genome **coverage**
- **Read length**
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- Genome **ploidy**

The practical will explore how combining these factors impacts the k-mer counts and spectra

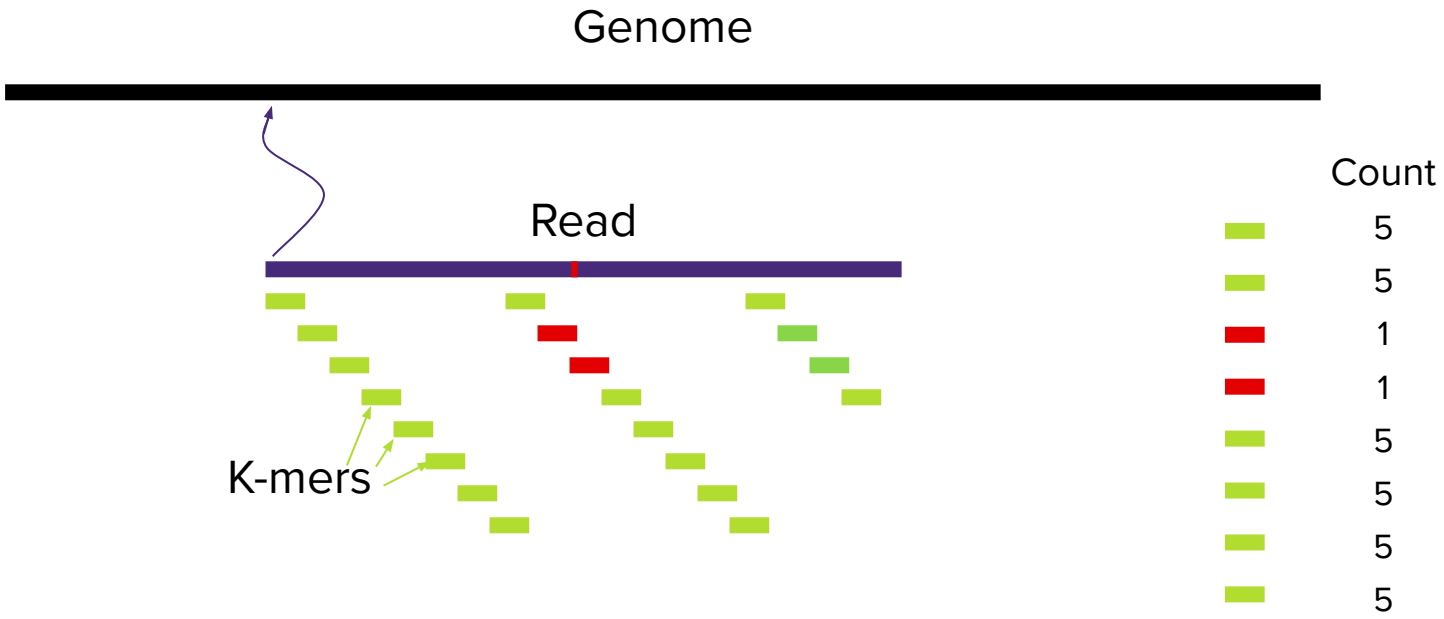
Factors that influence read k-mer counts : Errors



Oxford Nanopore reads: ~99.26% accurate
Pacbio HiFi reads: ~99.95% accurate
Short read SBS reads: ~99.92% accurate

https://www.pacb.com/technology/hifi-sequencing/#:~:text=PacBio%20is%20the%20only%20sequencing,address%20your%20toughest%20biological%20questions.

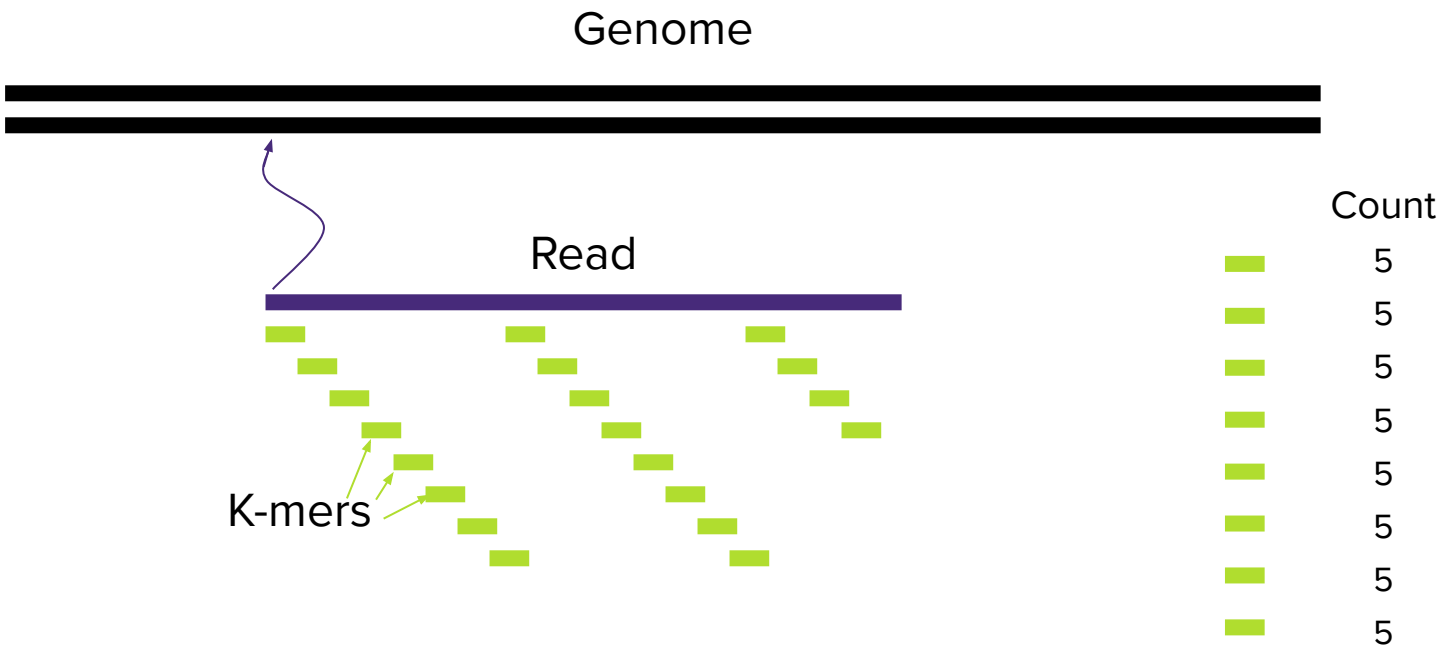
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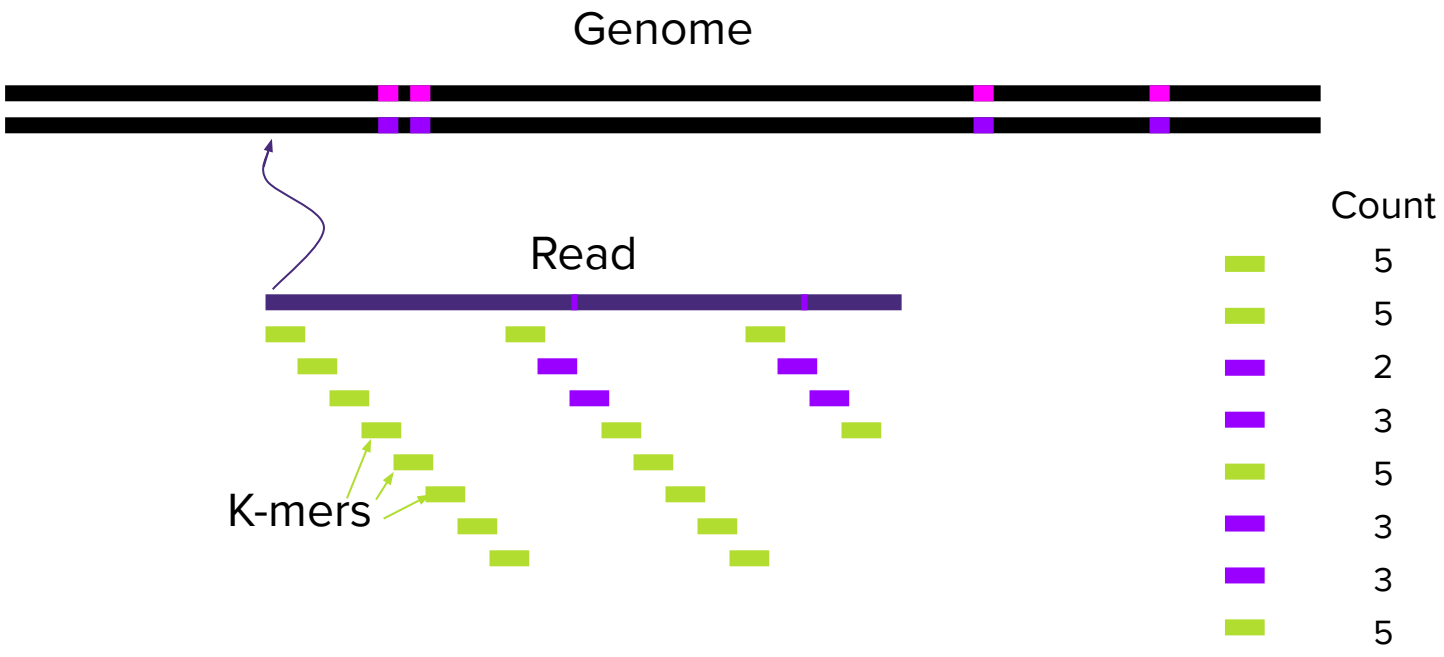
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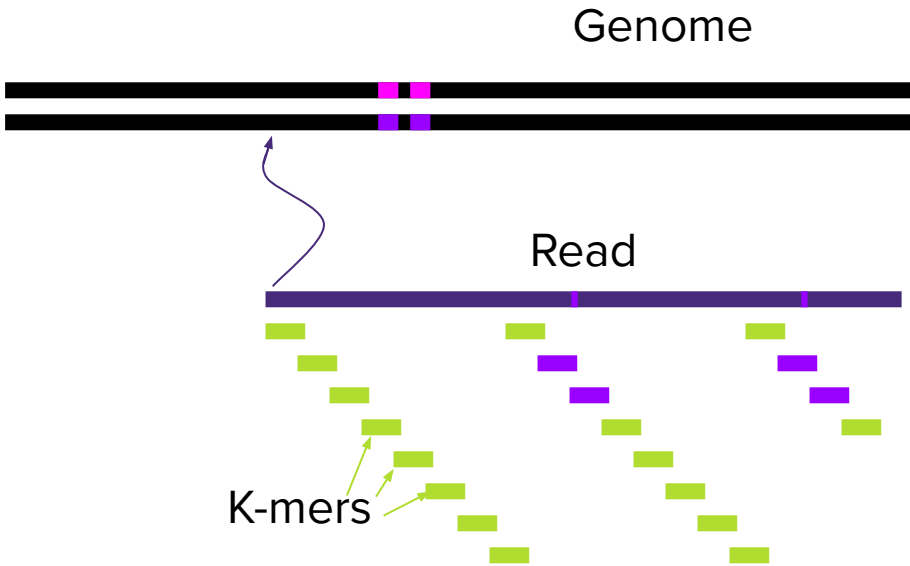
Factors that influence read k-mer counts : Heterozygosity

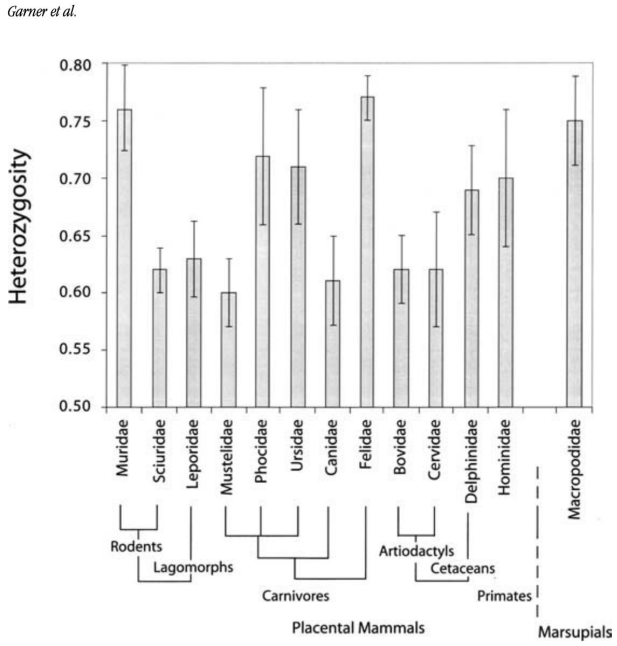


Factors that influence read k-mer counts : Heterozygosity



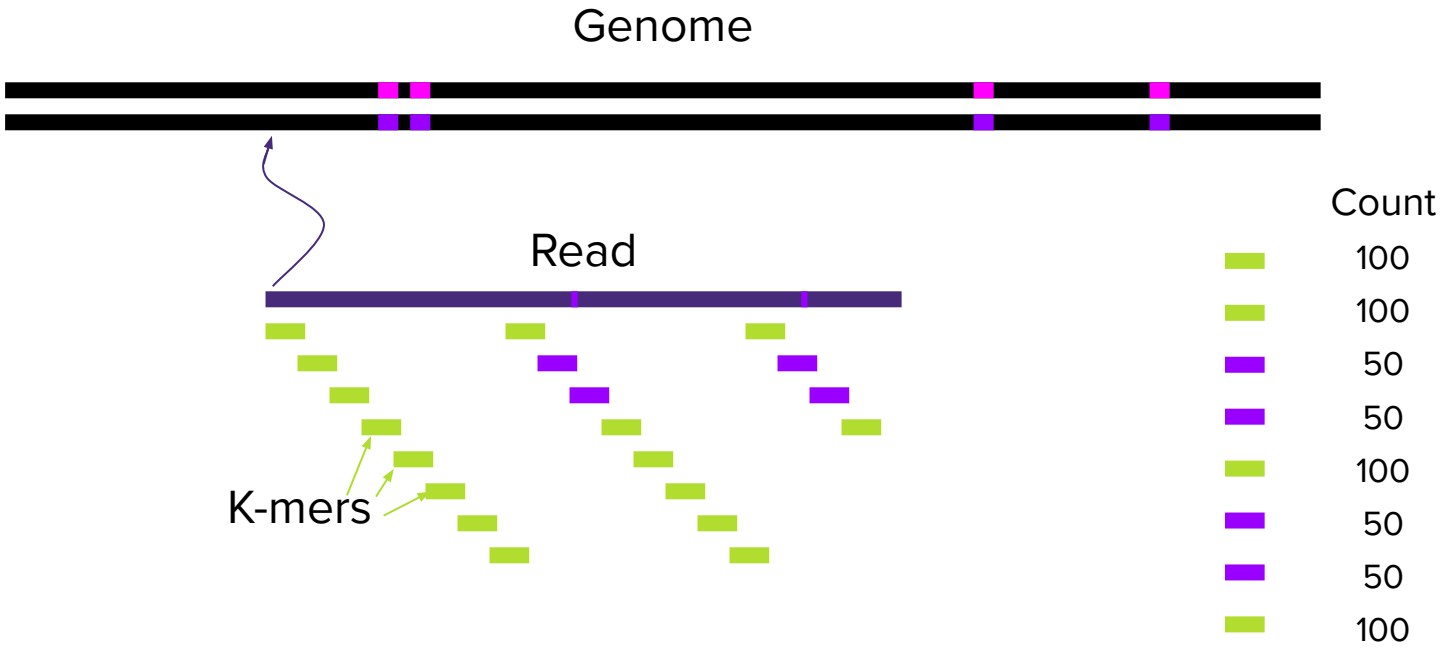
Factors that influence read k-mer counts : Heterozygosity





Garner et al. 2005. Patterns of Genetic Diversity and Its Loss in Mammalian Populations

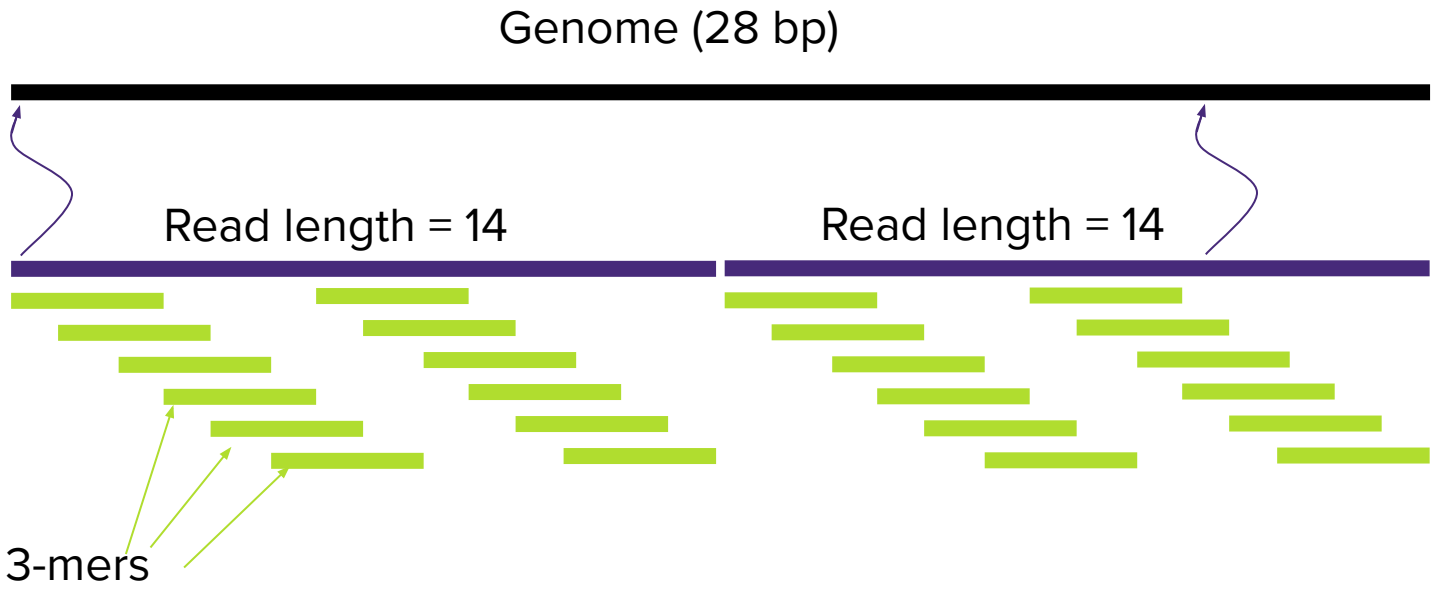
Factors that influence read k-mer counts : Coverage



Factors that influence read k-mer counts : Read length

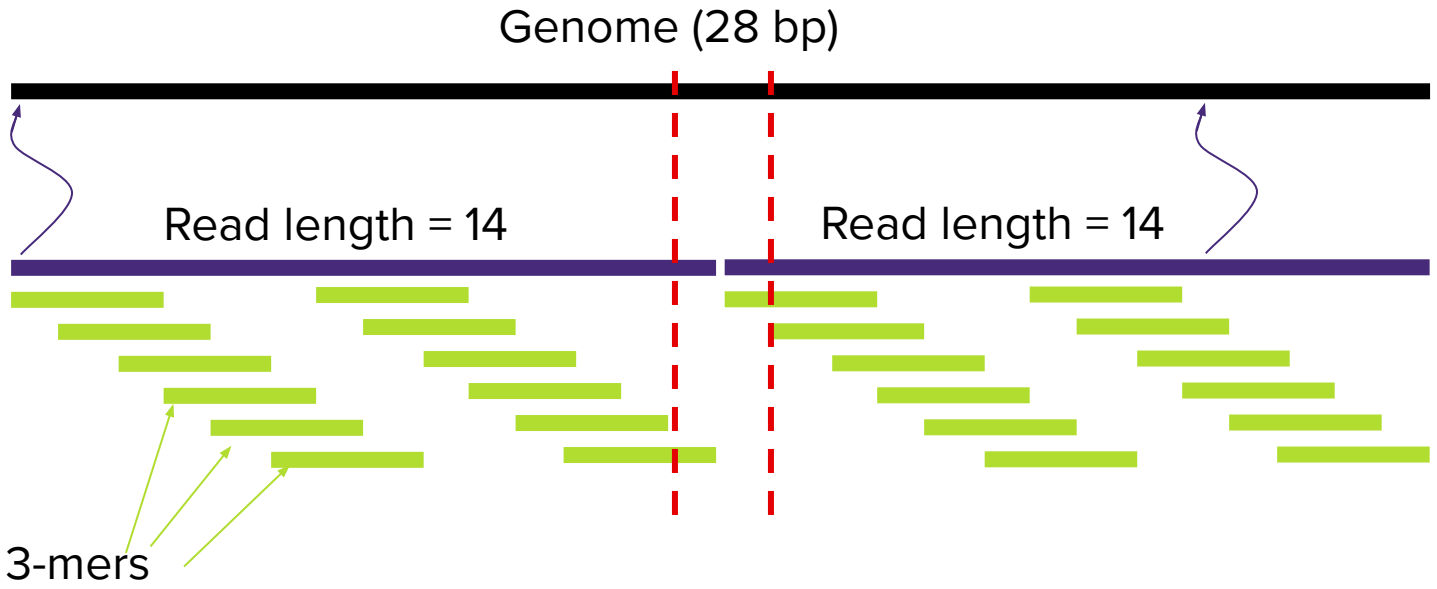


Factors that influence read k-mer counts : Read length



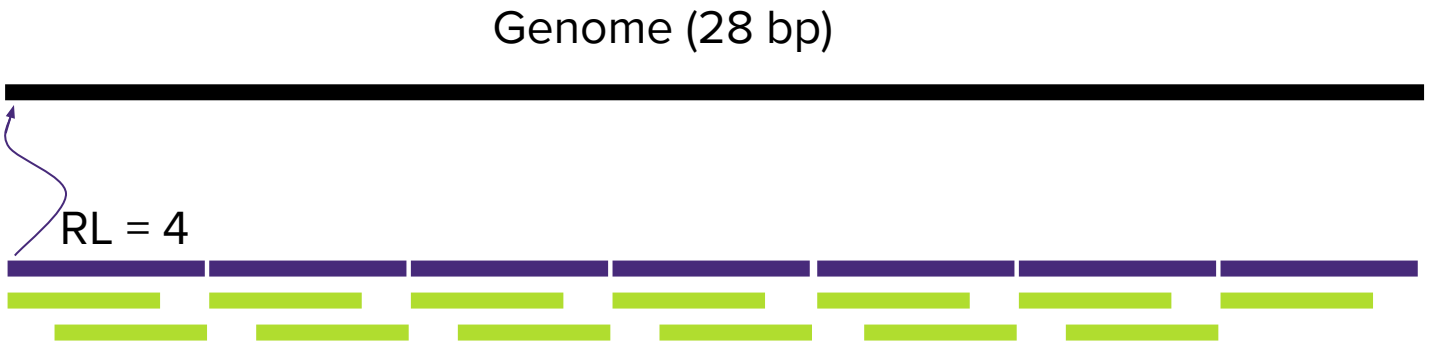
12 3-mers covers 1 read
1x read coverage of the genome requires 2 reads
But the k-mer coverage of the genome would be just under 3x (24 kmers / 28 bp)

Factors that influence read k-mer counts : Read length



12 3-mers covers 1 read
1x read coverage of the genome requires 2 reads
But the k-mer coverage of the genome would be just under 3x (24 kmers / 28 bp = ~0.86X)

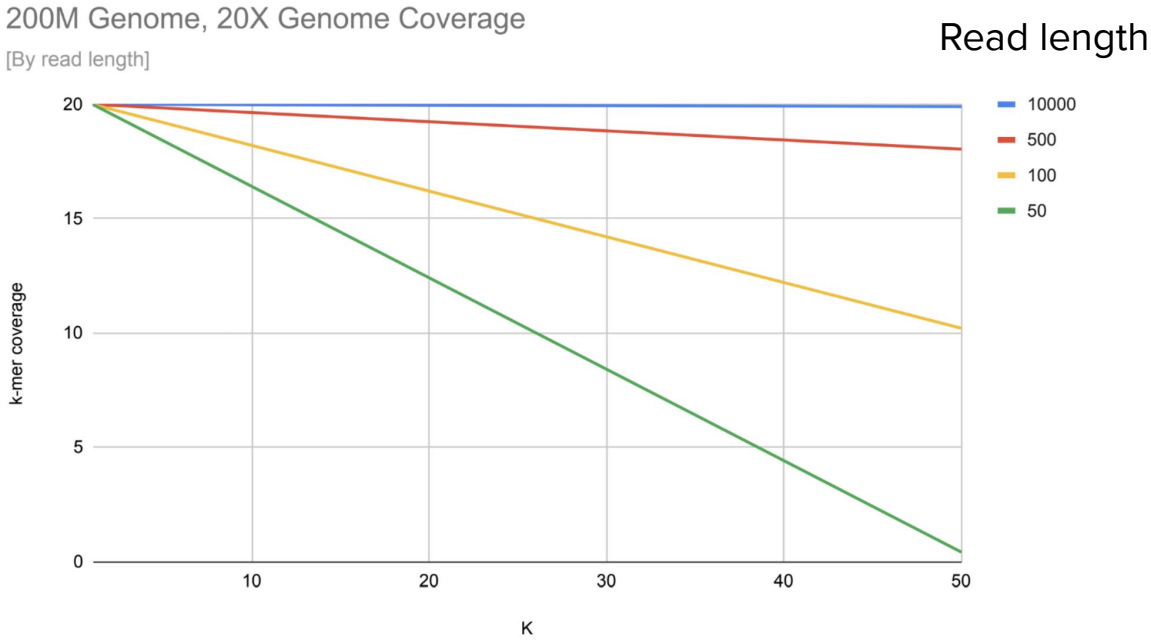
Factors that influence read k-mer counts : Read length



3-mers

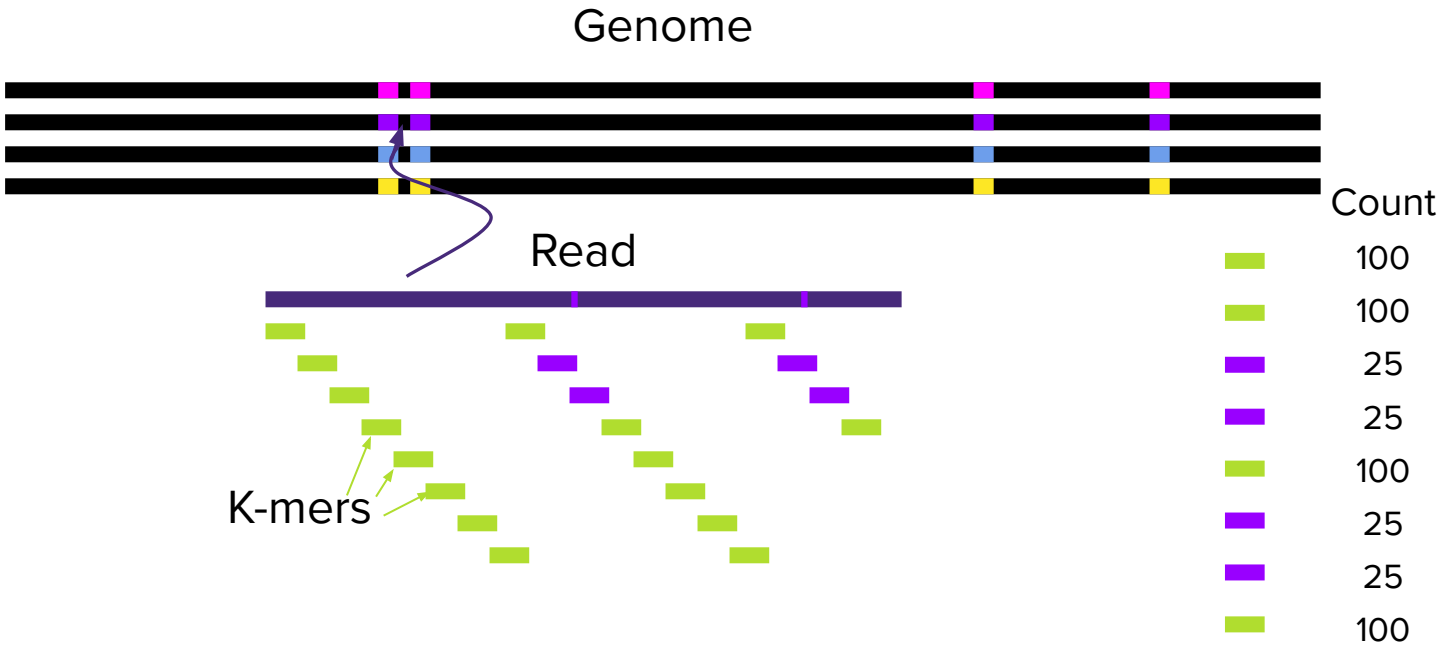
2 3-mers covers 1 read
1x read coverage of the genome requires 7 reads
But the k-mer coverage of the genome would be just 2x (14 kmers / 28 bp) for the same read coverage

Factors that influence read k-mer counts : Read length

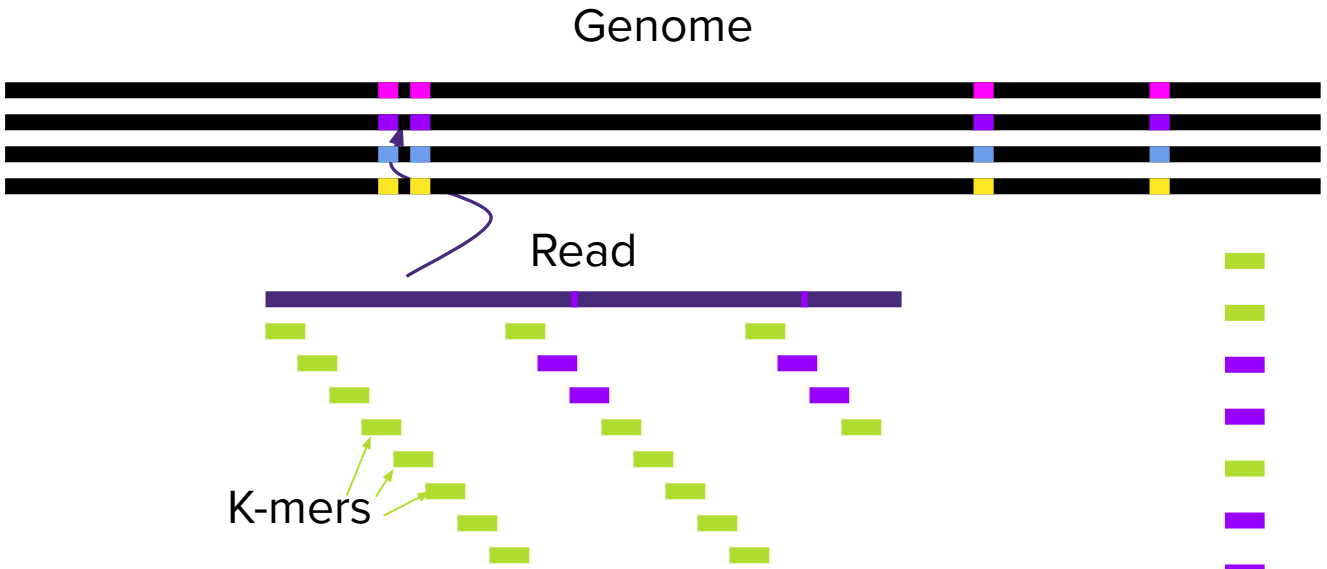


Supplemental Fig. S4: The relation of the *k*-mer coverage and *k* in relation to the read length

Factors that influence read k-mer counts : Ploidy



Factors that influence read k-mer counts : Ploidy

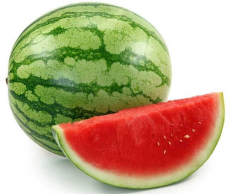


Count
100
100
25
25
100
25
25
100

Humans: 2x
Strawberry: 8x
Watermelon: 3x (sometimes)
Some frogs: 12x
Redwood: 6x









Outline

- K-mer review
- **Build a k-mer counter**
- de Bruijn graph intro
- Build a de Bruijn graph